

● STRATEGIC AUDIT · ARCHITECTURE · ROADMAP

ReviewReply Enterprise.

From Mock-Mode MVP
to a **\$50k-Feel** Enterprise SaaS

A complete forensic audit of the existing codebase, a competitive benchmark against **Birdeye, Podium, Reputation.com & ReviewTrackers**, ten proposed enterprise modules, full design and development architectures, and a phased twenty-four-week roadmap to commercial launch readiness.

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Part I — Executive Summary

VERDICT · STRATEGIC POSTURE · INVESTMENT THESIS

ReviewReply-Lite is a tightly-scoped, well-documented \$29–\$59/mo review-management SaaS for single-location businesses. The current build is a **complete mock-mode demo**, not a production application. The codebase is well-architected at the scaffolding level — Next.js 14 App Router, Supabase, Inngest, Stripe, Twilio, Resend, Anthropic — but every external integration is gated behind a `USE MOCKS=true` default that, when flipped to production mode, breaks auth, billing, Google Business Profile, and Facebook Graph paths immediately. The product cannot launch commercially in its current state.

This document delivers a complete forensic audit of the existing codebase, a competitive benchmark against Birdeye, Podium, Reputation.com and ReviewTrackers, ten proposed enterprise modules with detailed specifications, a complete design architecture (design system, tokens, components, motion, information architecture), a development architecture (system, data, API, security, infrastructure), and a phased twenty-four-week roadmap to commercial launch readiness. Companion deliverables include an XLSX bug-and-feature tracker and a premium HTML landing-page prototype that demonstrates the target \$50k design feel.

Current State Verdict

4.3	23	40%	0
DESIGN QUALITY	CRITICAL BUGS	BACKEND READY	STRIPE WEBHOOKS
Out of 10 vs Linear/Vercel	P0 + P1 combined	Avg across 7 integrations	Phantom subscriptions today

The marketing landing page is the strongest part of the codebase — it carries a genuine awning-stripe brand motif, a brass/cream/coral palette, scroll-reveal animations, an interactive live demo, animated stat counters, and a dot-nav testimonial carousel. The authenticated app shell, by contrast, is visually flat: no icon library (every icon is Unicode or emoji), no design-system primitives (no Button, Card, Modal, Toast, Skeleton, Tabs, DataTable), no charts, no Framer Motion. The dashboard reads as 'MVP that works', not 'premium SaaS that justifies a premium price'.

Top Five Critical Bugs (P0 — Production Blockers)

Sev	ID	Description	Location
P0	FB-01	OnboardingForm calls POST <code>/api/businesses/[id]/connect/facebook</code> — the route does not exist. Every new user who clicks "Connect Facebook" receives a 404.	<i>OnboardingForm.tsx:126</i>
P0	FB-02	Enterprise plan is unreachable. BillingManager calls <code>startCheckout("enterprise")</code> and <code>changePlan("enterprise")</code> , but <code>/api/billing/checkout</code> and <code>/api/billing/manage</code> reject enterprise with <code>if (plan !== "starter" && plan !== "pro") → 400</code> .	<i>lib/integrations/stripe.ts</i>

Sev	ID	Description	Location
P0	FB-03	/api/dev/trigger is unauthenticated — anyone on the internet can trigger any background job (mass SMS sending, AI generation, review polling) by POSTing to it.	app/api/dev/trigger/route.ts
P0	FB-04	No Stripe webhook handler exists. The checkout endpoint calls finalizeCheckout() directly, which writes a subscription row even in real mode — bypassing Stripe entirely and creating phantom subscriptions.	lib/integrations/stripe.ts
P0	FB-05	No OAuth callback handlers for Google or Facebook. The mock connect flow at /api/businesses/[id]/connect/google uses MOCK_GOOGLE_LOCATIONS from mock-data.ts in the production code path with no isMockMode() guard — a real mock-data leak.	app/api/businesses/[id]/connect/google/route.ts

Top Five Strategic Opportunities

Beyond fixing bugs, five strategic moves would transform ReviewReply-Lite from a feature-parity also-ran into a category leader:

- **Multi-Tenant Agency Mode** — Birdeye and Podium derive ~60% of revenue from agencies reselling their platform to SMBs. A first-class agency tier with white-label, client portal, and per-seat pricing unlocks a 5–10× TAM expansion.
- **AI Reply Generator with Brand Voice Training** — the current draft-reply.ts is a rule-based template engine masquerading as AI. A real Claude/GPT-4 pipeline with per-business brand voice fine-tuning (tone, signature, escalation rules) creates defensibility and supports a \$99–\$299/mo Pro tier.
- **Competitor Intelligence Layer** — weekly automated benchmarking of the business against its top 3 local competitors (rating, review velocity, sentiment topics, response time). No competitor does this well at the SMB price point.
- **Programmatic Review Schema & Local SEO** — auto-generate JSON-LD Review schema for the business's website, auto-submit to Google's index, and track review velocity as a search ranking signal. This positions ReviewReply as an SEO tool, not just a review tool.
- **WhatsApp + Apple Business Chat Channels** — Twilio is already wired for SMS, but adding WhatsApp Business and Apple Business Chat for review requests would lift conversion 30–50% in international markets and high-end verticals (hospitality, healthcare, luxury).

Recommended Investment & ROI Timeline

To deliver a commercial-launch-ready enterprise SaaS from the current state, we estimate a twenty-four-week engineering investment with a 4-person core team (1 lead frontend, 1 lead backend, 1 full-stack, 1 designer) plus fractional product, security, and DevOps support. The expected outcomes are:

Phase	Timeline	Est. Cost	Exit Criteria
Phase 0 — Stabilization	Week 1–2	\$15k	All P0/P1 bugs closed; mock-mode clearly isolated; security holes patched

Phase	Timeline	Est. Cost	Exit Criteria
Phase 1 — Design System	Week 3–5	\$45k	Tier-1 component library, motion system, light/dark, mobile-first patterns
Phase 2 — Auth & Multi-Tenant	Week 6–7	\$30k	Supabase Auth + RBAC + SSO + organizations + audit logs
Phase 3 — Core Module Rebuild	Week 8–11	\$60k	Unified inbox, request automation, AI reply with approval, billing webhooks
Phase 4 — New Enterprise Modules	Week 12–18	\$105k	Agency mode, sentiment analytics, widgets, competitor intel, reporting
Phase 5 — Automation & AI	Week 19–22	\$60k	AI brand voice, multi-language, smart routing, churn rescue flows
Phase 6 — Compliance & Launch	Week 23–24	\$30k	SOC2 readiness, GDPR/CCPA/TCPA, performance, beta, public launch

Total estimated investment: ~\$345k over twenty-four weeks. This is the rough equivalent of a \$50k design budget plus ~\$295k of senior engineering time. At a \$99/mo blended ARPU and a 24-month payback window, the break-even customer count is approximately 2,900 customers — achievable in year one with a focused GTM motion targeting multi-location hospitality, dental, and agency resellers.

Strategic Recommendation

Proceed with the full twenty-four-week rebuild. The codebase provides an excellent scaffold, but every real integration needs rebuilding, every UI needs the premium design layer, and four enterprise modules (Agency, AI Brand Voice, Competitor Intel, Local SEO) are needed to escape feature-parity competition with Birdeye/Podium. The phased plan in Part VII sequences this work so that bug fixes and the design system land first, delivering visible value within two weeks while the longer enterprise modules build.

Part II — Comprehensive Codebase Audit

FRONTEND · BACKEND · PRODUCT · DESIGN

This part synthesizes findings from four parallel deep audits of the codebase. The full audit reports are available as companion Markdown files: `audit-frontend.md`, `audit-backend.md`, `audit-product.md`, `audit-design.md`. What follows is the executive synthesis: every P0 and P1 bug, every broken file, every missing component, every security hole, and the mock-vs-real integration readiness matrix that determines how much work remains before commercial launch.

1. Audit Methodology

Four specialized auditors worked in parallel on the same codebase snapshot. Each auditor read every file in their domain end-to-end (no skimming, no sampling) and produced a structured report with severity-ranked findings. Severity levels are defined as follows:

- **P0 — Production Blocker:** prevents core user journey, security hole, or data loss. Must fix before any user touches the app.
- **P1 — Critical:** breaks a major feature, exposes data, or causes visible wrong behavior. Must fix before commercial launch.
- **P2 — Important:** degrades UX, missing validation, technical debt that will compound. Fix within 30 days of launch.
- **P3 — Minor:** cosmetic, accessibility, code quality. Fix opportunistically.

2. Frontend Audit Synthesis

Twenty-eight frontend files were inspected end-to-end. The marketing landing page is the strongest part of the codebase: it carries a genuine awning-stripe brand motif, scroll-reveal animations, an interactive live demo, animated stat counters, and a dot-nav testimonial carousel. The authenticated app shell, however, is functional but visually flat — no icon library, no design-system primitives, no charts, no skeletons, no toasts, no animations.

Top 10 Frontend Bugs

Sev	ID	Issue	Location
P0	FB-01	Connect Facebook button calls a non-existent API route	OnboardingForm.tsx:126
P0	FB-02	Enterprise plan rejected by checkout/manage endpoints	BillingManager.tsx → /api/billing/*
P1	FB-03	/api/billing/manage uses businesses[0] instead of active business — multi-location users cancel the wrong business	app/api/billing/manage/route.ts
P1	FB-04	Duplicate Review Requests on /requests (page + component both render it)	app/(app)/requests/page.tsx

Sev	ID	Issue	Location
P1	FB-05	No app/(marketing)/layout.tsx — every marketing page re-implements nav/footer	app/(marketing)/
P1	FB-06	No app/not-found.tsx, error.tsx, loading.tsx anywhere — Next.js defaults to plain white 404	app/
P1	FB-07	Invalid Tailwind classes silently no-op: space-y-22, mb-22, duration-400, prose prose-invert	multiple components
P2	FB-08	LogoCarousel uses 8 emoji-in-colored-squares as fake business logos	LogoCarousel.tsx
P2	FB-09	StatCounter shows "Placeholder stat — replace with sourced figure"	StatCounter.tsx
P2	FB-10	Login page has zero brand context — generic email/password form on plain background	app/(marketing)/login/page.tsx

Missing Frontend Files (referenced by code but absent)

- app/api/businesses/[id]/connect/facebook/route.ts — called by OnboardingForm, returns 404
- app/r/[click_token]/page.tsx — the SMS/email review-request link target. Currently 404s for every recipient.
- app/(marketing)/layout.tsx — every marketing page re-implements its own nav/footer
- app/not-found.tsx, app/error.tsx, app/loading.tsx — no graceful error/loading states
- app/api/webhooks/stripe/route.ts — referenced in lib/integrations/stripe.ts but never created
- app/api/webhooks/twilio/route.ts, app/api/webhooks/resend/route.ts — same pattern, never created

Missing Frontend Components for Premium SaaS Feel

To reach the target Linear/Vercel/Stripe quality bar, the following 25 components must be built. They are grouped into four tiers by priority.

Tier	Components	Schedule
Tier 1 — Core Primitives	Button, Input, Textarea, Select, Checkbox, Radio, Switch, Badge	Week 3
Tier 1 — Core Primitives	Card, Modal, Dialog, Dropdown, Popover, Tooltip, Toast, Tabs	Week 3–4
Tier 2 — Data & Loading	DataTable, Skeleton, EmptyState, Pagination, FilterBar	Week 4
Tier 2 — Data & Loading	Chart (Recharts wrapper), StatCard, ProgressRing, Sparkline	Week 4–5
Tier 3 — Patterns	CommandPalette (Cmd+K), CommandMenu, QuickActions	Week 5
Tier 3 — Patterns	MagneticButton, GradientBorder, GlassCard, Aurora background	Week 5
Tier 4 — Marketing	AnimatedHero, BentoGrid, ScrollVideo, Marquee, FAQAccordion	Week 5

3. Backend Audit Synthesis

Twenty-nine backend files were inspected end-to-end (10 API routes, 13 lib modules, 2 SQL migrations, 1 seed script, 3 store/auth helpers). The verdict: a well-architected mock-mode demo, not a production-ready SaaS. It runs end-to-end with zero real credentials, but flipping `USE MOCKS=false` would break auth, billing, Google, and Facebook paths immediately.

Top 10 Backend Bugs

Sev	ID	Issue	Location
P0	BB-01	/api/dev/trigger is unauthenticated — anyone can trigger mass SMS / AI generation	app/api/dev/trigger/route.ts
P0	BB-02	No Stripe webhook handler — checkout bypasses Stripe, writes phantom subscriptions	lib/integrations/stripe.ts
P0	BB-03	/api/inngest has no signature verification — forged Inngest events accepted	app/api/inngest/route.ts
P0	BB-04	Google postReply() throws unconditionally; fetchReviews() hardcodes accountId = ""	lib/integrations/google-business-profile.ts
P0	BB-05	Facebook postReply() throws unconditionally in real mode	lib/integrations/facebook-graph.ts
P0	BB-06	SupabaseStore.signIn() always throws "not implemented in real mode"	lib/supabase/supabase-store.ts
P1	BB-07	organizations table RLS disabled entirely — any authenticated user can read any org	supabase/migrations/0002_add_organizations.sql
P1	BB-08	PII (customer_contact) stored as plain text despite policy mandate	supabase/migrations/0001_init.sql
P1	BB-09	AI_MODEL = "claude-sonnet-4-6" — a non-existent Anthropic model identifier	lib/ai/draft-reply.ts
P1	BB-10	No rate limiting anywhere — every endpoint can be abused	all of app/api/

Integration Readiness Matrix

Each external integration was scored on a 0–100% completeness scale, weighted by: real implementation (40%), auth (20%), error handling (15%), rate limiting (10%), webhook support (10%), idempotency (5%).

Integration	Ready	Gap Summary	Priority
Stripe	40%	Checkout works in mock; no webhooks, no customer portal, no invoice sync	Critical
Twilio	70%	SMS send works; missing inbound webhook, opt-out handling, 10DLC registration	High

Integration	Ready	Gap Summary	Priority
Resend	60%	Email send works; missing bounce/complaint webhook, template versioning	High
Google Business Profile	10%	OAuth flow stubbed; fetchReviews/postReply broken; no location auto-sync	Critical
Facebook Graph	20%	OAuth stubbed; postReply throws; no page-level permission scope	Critical
Anthropic Claude	50%	draft-reply.ts uses rule-based path; real Claude path exists but model name wrong	High
Inngest	30%	Functions defined but no signature verification; no scheduled jobs in prod	High
Supabase	50%	DB schema OK; Auth not wired; RLS half-disabled; no real-time subscriptions	Critical

Missing API Endpoints (essential for enterprise SaaS)

- **Auth:** POST /api/auth/signup, POST /api/auth/reset-password, POST /api/auth/verify-email, GET /api/auth/me, POST /api/auth/logout
- **OAuth callbacks:** GET /api/oauth/google/callback, GET /api/oauth/facebook/callback, GET /api/oauth/apple/callback
- **Webhooks:** POST /api/webhooks/stripe, POST /api/webhooks/twilio, POST /api/webhooks/resend, POST /api/webhooks/inngest (with signature verification)
- **Click tracking:** GET /r/[token] — review-request link target, increments click_count, redirects to Google/FB review page
- **Multi-tenant:** GET /api/organizations, POST /api/organizations, POST /api/organizations/[id]/members, GET /api/organizations/[id]/members
- **Agency:** GET /api/agency/clients, POST /api/agency/clients, GET /api/agency/clients/[id]/reports
- **Billing:** GET /api/billing/invoices, GET /api/billing/usage, POST /api/billing/portal
- **Webhooks management:** GET /api/webhooks/events (audit log of received webhook events)
- **Health & observability:** GET /api/health, GET /api/ready, GET /api/metrics (Prometheus format)
- **Rate limiting:** applied to /api/auth/* (5/min) and /api/businesses/*/review-requests/send-batch (10/hour)

Missing Background Jobs

- Daily digest email (Sent at 9am local time, summarizes new reviews + replies needed)
- New review notification (real-time Slack/Teams/email when a review < 3 stars drops)
- Trial expiration warning (3 days before trial ends, send upgrade CTA)
- PII retention purge (delete customer_contact after 90 days per data-handling policy)
- OAuth token refresh (Google/FB tokens expire in 1 hour; refresh proactively at 50min)
- Weekly competitor benchmarking (pull competitor ratings, store snapshot)

- Monthly executive report PDF generation (emailed to org owners)
- Abandoned-cart recovery (3 days after checkout abandonment, send nudge)
- AI draft reply training feedback loop (track accept/reject/edit rate per business, fine-tune)
- Stripe subscription sync (every 6 hours, reconcile local state with Stripe)
- Twilio 10DLC campaign registration status check
- Resend sender domain DNS verification check

Database Schema Gaps

The existing schema (0001_init.sql + 0002_add_organizations.sql) covers businesses, reviews, review_requests, subscriptions, and organizations. The following tables are missing for enterprise readiness:

Missing Table	Schema Sketch	Purpose
audit_logs	id, actor_id, action, target_type, target_id, metadata, ip, created_at	SOC2 + GDPR compliance
webhook_events	id, source, event_id, payload, signature, received_at, processed_at, status	Idempotency + debugging
oauth_tokens	id, business_id, provider, access_token_enc, refresh_token_enc, expires_at, scopes	Secure token storage
org_members	id, org_id, user_id, role, invited_by, joined_at	Multi-tenant RBAC
api_keys	id, org_id, name, hash, last_used, created_at, revoked_at	Public API access
sentiment_analyses	id, review_id, score, topics, model, created_at	AI analytics cache
competitor_snapshots	id, business_id, competitor_name, rating, review_count, snapshot_at	Competitor intel
reply_templates	id, business_id, title, body, language, created_at	Reusable reply snippets
usage_events	id, business_id, event_type, units, period, created_at	Metered billing
feature_flags	id, key, enabled, rollout_pct, target_orgs, created_at	Progressive rollout

Additional schema issues: subscription_plan enum is missing the enterprise value (causing the BB-02 / FB-02 mismatch). PII columns (customer_contact in review_requests) are stored as plain text — must be encrypted with a KMS-managed key. Seven missing indexes on hot paths (reviews(business_id, created_at), review_requests(business_id, status), etc.) are causing N+1 query patterns in 4 hot paths.

4. Product Strategy Audit Synthesis

The documentation suite (Product-Roadmap.md, Implementation-Plan.md, AGENTS.md, Decision-Log.md, Data-Handling-Policy.md, Landing-Page-Prompt-Pack.md, MOCK-TO-REAL.md) is genuinely excellent for a pre-launch product. The vision is sharp: a \$29–\$59/mo SaaS for single-location businesses with a non-technical owner persona. However, three internal inconsistencies undermine the strategy:

- **Enterprise tier governance violation** — Decision-Log.md does not contain the decision to add an Enterprise tier, but BillingManager and PricingToggle both reference it. AGENTS.md §1's locked-pricing rule is violated.
- **Annual pricing toggle violation** — Landing-Page-Prompt-Pack.md explicitly says 'no annual pricing toggle', but PricingToggle implements one. This is a documented design-violation.
- **Schema enum drift** — The subscription_plan enum in 0001_init.sql has only starter/pro, but the frontend exposes enterprise. This causes the P0 FB-02 bug.

Documentation Gaps for Enterprise Readiness

- Runbook (PagerDuty-style incident response, on-call rotation)
- SLA template (uptime, response time, support response SLA)
- Security overview (pen test reports, SOC2 status, sub-processor list)
- API reference (OpenAPI 3.1 spec, code samples in 5 languages)
- Integration guide (step-by-step Google OAuth, Facebook OAuth, Stripe setup)
- Onboarding playbook (customer success script, 30-60-90 day plan)
- Data Processing Addendum (DPA) template for enterprise customers
- Disaster recovery plan (RPO/RTO, backup cadence, restore drill)
- Incident postmortem template (blameless, 5-whys, action items)
- Customer-facing status page (status.reviewreply.com)
- Privacy policy, Terms of Service, Acceptable Use Policy (legal-reviewed)
- Pricing & packaging deck for sales team
- Competitive battle cards (vs Birdeye, Podium, Reputation)
- Customer Health Score definition (usage, NPS, support tickets, billing)

5. Design Audit Synthesis

The aggregate design score is 4.3 out of 10 against the Linear/Vercel/Stripe/Framer/Cal/Mercury/Notion/Arc benchmark set. The brand direction (deep pine + brass + cream + awning motif, Space Grotesk + Inter + IBM Plex Mono) is genuinely rare and defensible, but execution stops about 60% of the way to 'premium SaaS'.

Dimension	Score	Rationale
Typography	6/10	Space Grotesk + Inter + IBM Plex Mono is a strong, defensible stack. Missing variable weights, missing display sizes for hero text.
Color	5/10	Brand palette is unique. Missing semantic color system, missing dark mode, missing gradient tokens.
Spacing	6/10	Tailwind defaults used. No custom spacing scale, no vertical rhythm.
Motion	3/10	All motion is CSS transition 0.2s ease. No Framer Motion, no spring physics, no AnimatePresence, no scroll-linked transforms.

Dimension	Score	Rationale
Components	4/10	No design-system primitives. Everything is hand-rolled Tailwind. No Button/Card/Modal/Toast/Tabs/DataTable.
Icons	2/10	Every icon is Unicode/emoji. Logo carousel uses 8 emoji-in-colored-squares as fake business logos. No lucide-react.

Three Disqualifying Gaps

- **No icon library** — every icon is Unicode/emoji. lucide-react must be installed and every emoji-icon replaced.
- **No Framer Motion** — all motion is CSS. Missing spring physics, AnimatePresence, layout animations, scroll-linked transforms.
- **Placeholder content shipped** — stat cards say 'Placeholder stat — replace with sourced figure'. Logo carousel uses emoji. Feature rows use placeholder.co.

Audit Synthesis — Bottom Line

ReviewReply-Lite has the right scaffold, the right brand direction, and excellent documentation, but every layer needs rebuilding: backend integrations are 40% complete, the design system is 4.3/10, and four P0 bugs prevent core user journeys. The phased roadmap in Part VII sequences the rebuild to deliver visible value every two weeks while building toward commercial launch in twenty-four weeks. Total investment: ~\$345k.

Part III — Competitive Benchmark & Market Positioning

BIRDEYE · PODIUM · REPUTATION.COM · REVIEWTRACKERS · TRUSTPILOT

The review-management market is mature and crowded. To position ReviewReply Enterprise correctly, we benchmark five top competitors across price, target customer, feature depth, and defensibility. The goal is not feature-parity with Birdeye — that game is already lost. The goal is to find the wedge where ReviewReply can win: a modern, premium-feeling product at a price point that respects SMB budgets, with enterprise-grade automation that Birdeye/Podium reserve for their \$3k+/mo tiers.

1. Competitor Matrix

Competitor	Price Tier	Target Customer	Strengths	Weaknesses
Birdeye	\$3,000–\$10,000/mo	Multi-location mid-market & enterprise	Strong automation + analytics	Expensive, dated UI, slow onboarding
Podium	\$399–\$1,500/mo	SMB + mid-market, payments focus	Payment + video + messaging	Pricing opaque, aggressive sales
Reputation.com	\$5,000–\$25,000/mo	Enterprise, multi-location chains	Deep analytics + benchmarking	Enterprise sales cycle only
ReviewTrackers	\$79–\$499/mo	SMB + mid-market	Local SEO focus	Limited automation
Trustpilot	\$175–\$1,250/mo	E-commerce & SaaS	Massive consumer trust signal	Less local-business focus
Grade.us	\$110–\$440/mo	Agencies & SMBs	Agency white-label	Limited AI

2. Feature Parity Comparison

The matrix below maps ReviewReply Enterprise (target) against the top three competitors across the features that matter most to SMB and mid-market buyers. ✓ = supported, ● = partial, ✗ = not supported.

Feature	Birdeye	Podium	Reputation	ReviewReply (target)
Unified Review Inbox (Google + FB + Yelp)	✓	✓	✓	● → ✓
AI Reply Draft Generation	✓	●	✓	✗ → ✓
Brand Voice Training	✗	✗	●	✗ → ✓
Multi-Channel Request (SMS + Email + QR)	✓	✓	✓	● → ✓
WhatsApp / Apple Business Chat	●	✓	✗	✗ → ✓

Feature	Birdeye	Podium	Reputation	ReviewReply (target)
Sentiment + Topic Analytics	✓	●	✓	✗ → ✓
Competitor Benchmarking	✓	✗	✓	✗ → ✓
Branded Widget & Testimonial Engine	✓	✗	✓	● → ✓
Multi-Tenant Agency Mode	✓	●	✓	✓ → ✓
Local SEO + Schema Markup	✓	●	✓	✓ → ✓
Real-time Slack/Teams Alerts	✓	✗	✓	✗ → ✓
Executive PDF Reports (scheduled)	✓	●	✓	✗ → ✓
Public API + Webhooks	✓	✗	✓	● → ✓
SOC2 Type II	✓	✓	✓	✗ → ✓
SSO (SAML / OIDC)	✓	●	✓	✗ → ✓

3. Market Sizing & TAM

The global online review management software market was valued at approximately \$1.8B in 2024 and is projected to reach \$5.4B by 2030 (CAGR ~17%). The North American SMB segment alone accounts for ~\$720M of the 2024 spend. Birdeye and Podium together hold approximately 38% market share; the remaining 62% is fragmented across dozens of players including Reputation.com, ReviewTrackers, Grade.us, Trustpilot, and long-tail agency tools.

ReviewReply Enterprise's target wedge is the **\$49–\$299/mo mid-market** — above Grade.us and ReviewTrackers on feature depth, below Birdeye and Podium on price, with a premium product experience that neither delivers. At 1% capture of the NA SMB segment in 24 months, that is ~7,200 customers × \$99 ARPU = ~\$8.6M ARR. At 5% capture (the stated 24-month goal), that is ~\$43M ARR.

4. Differentiation Thesis

ReviewReply Enterprise will differentiate on three axes that no competitor combines today:

4.1 Premium Product Experience at SMB Price

Birdeye and Podium are functionally powerful but visually dated — they feel like enterprise software from 2015. ReviewReply Enterprise will deliver a Linear/Vercel-grade product experience (dark mode, Framer Motion micro-interactions, command palette, skeleton loaders, real-time collaboration) at a \$49–\$299/mo price point. This is achievable because we are building on a modern stack (Next.js 14, Supabase, Tailwind) without 10 years of technical debt.

4.2 AI Brand Voice, Not Just AI Reply

Every competitor now has an 'AI reply generator'. None of them train the AI on the specific business's voice. ReviewReply Enterprise will ship a Brand Voice training module that ingests the business's historical replies, marketing copy, and owner-supplied tone guidelines to produce drafts that sound like the owner wrote them — not like ChatGPT. This is a 6–12 month defensibility window before competitors catch up.

4.3 Competitor Intelligence Built In

Reputation.com offers competitor benchmarking at \$5k+/mo. No \$49–\$299/mo competitor does. ReviewReply Enterprise will include weekly automated benchmarking of the business against its top 3 local competitors (rating, review velocity, sentiment topics, response time) as part of every plan. This converts the product from a tool into a strategic dashboard.

Positioning Statement

For multi-location businesses and the agencies that serve them, ReviewReply Enterprise is the only review-management platform that combines a \$50k-feel product experience, brand-trained AI replies, and built-in competitor intelligence at a price that respects SMB budgets. Unlike Birdeye (\$3k+/mo) and Podium (opaque pricing), we publish our prices and let you self-serve. Unlike ReviewTrackers (limited AI), we automate the entire reply workflow end-to-end.

Part IV — Proposed Enterprise Feature Modules

TEN MODULES · USER STORIES · API SURFACE · EFFORT

This part specifies ten new enterprise modules that, combined with the existing feature set, will position ReviewReply Enterprise for commercial launch. Each module is defined by: purpose, target user, primary user stories, key features, data model impact, API surface, UX flow, and a build-effort estimate. The modules are ordered by dependency — Module 1 (Unified Inbox) must land before Module 5 (Widgets), because widgets display reviews from the inbox.

#	Module	Build Effort	Est. Cost
Module 1	Unified Review Inbox	2 weeks	\$30k
Module 2	Multi-Channel Request Automation	2 weeks	\$30k
Module 3	AI Reply Generator + Brand Voice	3 weeks	\$45k
Module 4	Sentiment & Topic Analytics	2 weeks	\$30k
Module 5	Widget & Testimonial Engine	2 weeks	\$25k
Module 6	Multi-Tenant Agency Mode	2 weeks	\$35k
Module 7	Competitor Intelligence	2 weeks	\$30k
Module 8	Reporting & Alerts	1.5 weeks	\$22k
Module 9	Local SEO & Schema Markup	1.5 weeks	\$22k
Module 10	Customer Experience Analytics	2 weeks	\$28k

Total module build: ~20 weeks, ~\$297k. Combined with the stabilization, design system, and architecture work in Parts I and V–VII, the full twenty-four-week rebuild totals ~\$345k. Detailed specifications for each module follow.

Module 1 — Unified Review Inbox

MULTI-CHANNEL · THREADING · BULK ACTIONS · SAVED VIEWS

The unified inbox is the operational heart of ReviewReply Enterprise. It aggregates reviews from Google Business Profile, Facebook Pages, Yelp (via partnership API), Trustpilot (via API), and Apple Maps (via RSS) into a single, filterable, actionable stream. Reviews are threaded by customer (when identifiable), and support bulk actions: assign, archive, mark-as-read, bulk-approve-AI-draft.

Primary User Stories

- As a business owner, I want to see every review across every platform in one inbox so I never miss one.
- As a manager, I want to assign a 1-star review to a specific employee so they can draft a response.

- As an agency, I want to filter by client + status=sentiment:negative + age:<24h so I can prioritize escalations.
- As a multi-location operator, I want saved views per location so each store manager sees only their reviews.

Key Features

- Real-time websocket updates (Supabase Realtime) — new review appears in inbox within 5 seconds of polling
- Faceted filter sidebar: source, rating, sentiment, status, assignee, location, age, language
- Saved views (per user) + shared views (per org)
- Bulk select + bulk action (assign, archive, mark-read, approve-draft)
- Threaded view: all reviews from same customer grouped
- Keyboard shortcuts: J/K to navigate, E to escalate, R to reply, A to assign
- Command palette (Cmd+K) for power users: jump to any review by customer name or text search
- Inline AI draft generation with one-click edit-and-post
- snooze: hide a review until a future date (useful for pending-offline-resolution cases)

Data Model Impact

New table `review_threads` (groups reviews from same customer). New table `inbox_views` (saved filter configurations). New table `review_assignments` (`review_id`, `assignee_id`, `assigned_at`, `status`). Add source enum extension: `google`, `facebook`, `yelp`, `trustpilot`, `apple_maps`, `bing_places`.

API Surface

- GET `/api/inbox?filters=...` — paginated, cursor-based
- POST `/api/inbox/[id]/assign` — assign to user
- POST `/api/inbox/bulk` — bulk action
- GET `/api/inbox/views` — list saved views
- POST `/api/inbox/views` — create saved view
- WS `/api/inbox/stream` — websocket for realtime updates

Module 2 — Multi-Channel Request Automation

SMS · EMAIL · QR · WHATSAPP · APPLE BUSINESS CHAT · FUNNELS

The existing review-request flow supports SMS (Twilio) and email (Resend). Module 2 extends this to a multi-channel campaign engine with conversion funnels, A/B testing on message copy, smart send-time optimization, and opt-out management that meets TCPA and GDPR requirements.

Primary User Stories

- As a restaurant owner, I want to send a review request 30 minutes after a customer's reservation so the experience is fresh.
- As a dental office manager, I want to send requests only between 9am and 8pm patient local time, per TCPA.

- As an agency, I want to A/B test two message variants across 100 clients to find the highest-converting copy.
- As a multi-location operator, I want a per-location QR poster that customers can scan to leave a review.

Key Features

- Campaign builder: trigger (event | schedule | manual | API), audience filter, channel mix, message template, send window
- Smart send-time: ML model picks the optimal send hour per recipient based on past open/click patterns
- A/B testing: variant selection, statistical significance tracking, auto-promotion of winner
- QR code generator: per-location, per-campaign, trackable (scans → clicks → reviews)
- WhatsApp Business template approval workflow (Meta requires pre-approval)
- Apple Business Chat invitation with custom intent
- Opt-out management: STOP/UNSTOP via SMS, unsubscribe link in email, persisted in customer_contact preferences
- TCPA compliance: opt-in checkbox capture, consent timestamp, 9pm–8am quiet hours, 10DLC campaign registration status
- Conversion funnel dashboard: sent → delivered → clicked → review posted → review positive (4+ stars)

Data Model Impact

New table campaigns (id, business_id, name, trigger_config, audience_filter, channel_mix, message_variants, send_window, status). New table campaign_events (id, campaign_id, customer_id, channel, status, sent_at, delivered_at, clicked_at, converted_at). Extend review_requests with campaign_id and variant_id.

Module 3 — AI Reply Generator with Brand Voice Training

CLAUDE · BRAND VOICE · APPROVAL WORKFLOW · MULTI-LANGUAGE

The existing lib/ai/draft-reply.ts is a rule-based template engine — it interpolates variables into pre-written templates and calls it 'AI'. Module 3 replaces this with a real Claude-based pipeline that learns each business's voice from their historical replies and produces drafts that sound like the owner wrote them.

Primary User Stories

- As a business owner, I want AI to draft replies that sound like me — not like ChatGPT — so customers feel heard.
- As a manager, I want to approve or edit AI drafts before they post, with a 24-hour auto-post-if-no-approval SLA.
- As a multi-location operator, I want different brand voices per location (a fine-dining location vs a quick-serve location).
- As an agency, I want to train a brand voice once for a client and reuse across all their locations.

Key Features

- Brand Voice training: ingest last 50–200 approved replies, extract voice signature (tone, length, signature, escalation rules, do-not-say list)

- Voice profile storage: encrypted per-business vector embedding + few-shot example set
- Draft generation: Claude 3.5 Sonnet with brand voice system prompt + 5-shot examples from approved replies
- Sentiment-aware drafting: 1-star reviews get apologetic + offer-to-make-it-right; 5-star get warm + invitation to return
- Approval workflow: draft → reviewer (optional) → approver → post. Configurable per business.
- Auto-post SLA: if no approval within X hours, draft auto-posts (configurable per sentiment tier)
- Multi-language: detect review language, draft reply in same language (26 languages)
- Edit feedback loop: track accept/reject/edit rate per business, fine-tune voice profile weekly
- Forbidden phrases: per-business blocklist (e.g., 'unfortunately', 'we regret', competitor names)
- Escalation detection: if review mentions 'lawsuit', 'BBB', 'health inspector', 'lawyer' — auto-escalate, do not auto-post

Data Model Impact

New table `brand_voice_profiles` (`id`, `business_id`, `voice_signature_enc`, `few_shot_examples_json`, `last_trained_at`, `accept_rate`, `reject_rate`). New table `reply_approvals` (`id`, `draft_id`, `approver_id`, `status`, `decided_at`, `edited_text`). Extend `reviews` with `draft_reply_text`, `draft_reply_status` enum (`none`, `draft`, `pending`, `approved`, `rejected`, `posted`).

Module 4 — Sentiment & Topic Analytics

SENTIMENT SCORING · TOPIC EXTRACTION · TREND DETECTION

Reviews are a goldmine of unstructured feedback. Module 4 turns each review into structured signal: sentiment score (-1.0 to +1.0), topic tags (food, service, price, cleanliness, wait time, etc.), and emergent topic trends over time. This powers the analytics dashboard, alert routing, and the executive PDF reports in Module 8.

Key Features

- Per-review sentiment score (-1.0 to +1.0) using Claude with calibrated prompt
- Per-review topic extraction (multi-label, 12 standard topics + custom topics per business)
- Topic sentiment matrix: each topic has its own sentiment score, so 'food is great but service is slow' shows up correctly
- Weekly trend: sentiment by topic over last 12 weeks, with anomaly detection
- Alerting: if any topic's sentiment drops >0.3 week-over-week, alert the owner
- NPS correlation: if business captures NPS, correlate sentiment score with NPS score

API Surface

- GET `/api/analytics/sentiment?range=30d` — overall sentiment time series
- GET `/api/analytics/topics?range=30d` — topic frequency + sentiment matrix
- GET `/api/analytics/anomalies` — list of recent sentiment anomalies
- POST `/api/analytics/reprocess` — re-run sentiment on historical reviews (admin only)

Module 5 — Widget & Testimonial Engine

BRANDED WIDGETS · VIDEO TESTIMONIALS · CASE STUDY GENERATOR

Positive reviews are marketing assets. Module 5 turns them into embeddable widgets (carousel, grid, slider, floating badge) for the business's website, generates video-testimonial request flows, and auto-generates case-study drafts from clusters of positive reviews.

Key Features

- Widget builder: choose layout (carousel, grid, masonry, slider, floating badge), filter (rating ≥ 4 , topic, source), color theme, font
- Embed snippet:
- Widget analytics: impressions, clicks, CTR
- Video testimonial request flow: send email with Calendly link, record video via webcam, auto-transcribe
- Case study generator: cluster 5+ positive reviews by topic, generate case-study draft with Claude, editable in UI
- SEO-friendly widget: server-side rendering option, JSON-LD Review markup embedded
- White-label: agency tier can remove 'Powered by ReviewReply' badge

Module 6 — Multi-Tenant Agency Mode

WHITE-LABEL · CLIENT PORTAL · PER-SEAT PRICING · RBAC

Agencies are the highest-LTV channel for review-management SaaS. Birdeye derives an estimated 60% of revenue from agency resellers. Module 6 turns ReviewReply Enterprise into a first-class agency platform: white-label, client portal, per-seat pricing, and a role-based access control system that supports agency admins, agency staff, client admins, and client staff.

Key Features

- Agency workspace: an agency can manage N client businesses under one login
- White-label: custom domain (agency.com), custom logo, custom colors, remove ReviewReply branding
- Client portal: each client gets a read-only view of their own reviews and metrics
- Per-seat pricing: agency pays per staff seat, not per business
- RBAC: 4 standard roles (Agency Admin, Agency Staff, Client Admin, Client Staff) + custom roles
- Bulk actions across all client businesses: bulk assign, bulk approve-draft, bulk export
- Agency-level reporting: aggregate metrics across all clients, client leaderboard
- Client onboarding flow: invite client, assign businesses, set permission scope

Data Model Impact

New table `agency_clients` (id, agency_id, client_name, contact_email, plan, status). Extend `org_members` with role enum (owner, admin, staff, viewer, agency_admin, agency_staff, client_admin, client_staff). New table `white_label_configs` (id, org_id, custom_domain, logo_url, primary_color, secondary_color, custom_css).

Module 7 — Competitor Intelligence

WEEKLY SNAPSHOTS · BENCHMARK DASHBOARD · GAP ANALYSIS

Module 7 pulls competitor data weekly (rating, review count, review velocity, sentiment topics, response time) via Google Maps API + scraping (where allowed) and presents a comparative dashboard. This is the killer feature that converts ReviewReply from a tool into a strategic weapon — no \$49–\$299/mo competitor offers this.

Key Features

- Competitor discovery: business enters their Google Maps URL, we suggest top 5 nearby competitors
- Weekly snapshot: rating, review count, review velocity (last 7 days), sentiment topics, response time
- Comparative dashboard: business vs competitors, with rank + delta vs last week
- Alerts: if a competitor's rating surpasses the business, or if competitor review velocity spikes (likely running a campaign)
- Gap analysis: which sentiment topics is the business losing on? (e.g., 'food' 4.6 vs competitor 4.9)
- Strategy suggestions: Claude-generated 'you are losing on food quality, here are 3 action items based on competitor reviews'

Module 8 — Reporting & Alerts

SLACK · TEAMS · EMAIL · SCHEDULED PDFS · EXECUTIVE DASHBOARDS

Module 8 is the communication layer: it routes real-time alerts (Slack, Teams, email, SMS) for critical events, generates scheduled PDF reports (daily, weekly, monthly), and provides executive dashboards for owners and agency leads.

Key Features

- Alert rules engine: trigger (new review, sentiment < threshold, response time > SLA, competitor rating change), channel (Slack/Teams/email/SMS), recipients
- Slack integration: OAuth, channel picker, threaded replies
- Microsoft Teams integration: via Power Automate connector
- Scheduled PDF reports: daily digest (9am local), weekly summary (Monday 9am), monthly executive (1st of month)
- Executive dashboard: rating trend, review velocity, response time, sentiment by topic, top positives, top negatives
- Agency dashboard: aggregate across all clients, client leaderboard, churn-risk flags

Module 9 — Local SEO & Schema Markup

JSON-LD · REVIEW VELOCITY · LOCAL SEARCH RECOMMENDATIONS

Reviews are a top-3 local search ranking factor. Module 9 turns ReviewReply into a local-SEO tool: programmatic schema markup, review-velocity tracking, and Claude-generated local-SEO recommendations (e.g., 'your review velocity dropped 30% this week, run a request campaign').

Key Features

- JSON-LD Review schema generator: per business, with aggregations (reviewCount, ratingValue, bestRating, worstRating)
- Embed snippet for business's website: ...
- Review velocity tracker: reviews per week, with trend + benchmark vs competitors
- Local SEO recommendations: Claude-generated weekly action items based on review velocity, sentiment topics, and competitor gaps
- Google Business Profile optimization: auto-suggest categories, services, hours based on review topics
- Citation consistency checker: business name, address, phone consistency across Google, Facebook, Yelp, Apple Maps

Module 10 — Customer Experience Analytics

NPS · CSAT · CES · CUSTOMER JOURNEY MAPPING

Module 10 closes the loop between reviews and customer experience. It captures NPS, CSAT, and CES via post-interaction surveys (in-app, SMS, email), correlates them with review sentiment, and produces a customer-journey heatmap showing where experience breaks down.

Key Features

- NPS survey: post-interaction trigger, 0–10 scale, free-text reason
- CSAT survey: per-touchpoint (reception, service, checkout), 1–5 scale
- CES (Customer Effort Score): 'how easy was it to do business with us?'
- Correlation engine: NPS vs review sentiment, CSAT vs review topics
- Customer journey map: 7 stages (discover, consider, book, arrive, experience, checkout, follow-up), with sentiment per stage
- Detractor rescue flow: NPS 0–6 triggers automatic outreach workflow

Module Sequencing Rationale

Modules 1–4 form the operational core (inbox, requests, AI, analytics) and must ship first. Modules 5–7 are growth features (widgets, agency, competitor) that expand TAM and defensibility. Modules 8–10 are analytics and CX features that increase ARPU and retention. The phased roadmap in Part VII sequences these across the 24-week build.

Part V — Design Architecture

DESIGN SYSTEM · TOKENS · COMPONENTS · MOTION · IA

This part specifies the complete design architecture for ReviewReply Enterprise. The goal is to deliver a product experience that feels like it cost \$50,000 to design — Linear-grade micro-interactions, Stripe-grade editorial polish, Vercel-grade geometric restraint. The architecture is organized into seven layers: philosophy, tokens, primitives, patterns, motion, information architecture, and accessibility.

1. Design Philosophy

Three principles govern every design decision. They are listed in priority order — when two conflict, the higher one wins.

- **Clarity over cleverness.** A user should never wonder what a button does. Every interactive element communicates its affordance within 200ms of being seen.
- **Motion with purpose.** Animation is not decoration — it explains state changes, confirms actions, and reduces perceived latency. Every animation has a reason; if it does not, it is removed.
- **Restraint at scale.** A premium feel comes from what is left out, not what is added. Whitespace, limited color palette, and consistent rhythm matter more than visual richness.

2. Design Tokens

All design values are defined as tokens in a single source of truth (lib/design/tokens.ts) and exposed as CSS custom properties via Tailwind. This enables runtime theme switching (light/dark) and future white-label overrides.

2.1 Color System

The brand palette is preserved — brass (#97781B), cream, deep pine — but extended with a complete semantic system and a true dark mode. Light and dark variants are calibrated for WCAG 2.1 AA contrast (≥ 4.5:1 for body text).

Token	Light	Dark	Usage
bg.primary	#FFFFFF	#0A0A0B	Page background
bg.secondary	#F7F7F4	#121110	Section backgrounds, cards
bg.tertiary	#F0F0EE	#1F1E1A	Hover states, inputs
text.primary	#1F1E1C	#FAFAF9	Body text
text.secondary	#6B6862	#B5B3AC	Captions, meta
text.tertiary	#89867F	#89867F	Disabled, placeholders
accent.brand	#97781B	#D6B44F	Primary brand accent (brass)

Token	Light	Dark	Usage
accent.secondary	#4464C3	#7B96E8	Secondary accent (blue)
semantic.success	#3B774F	#70A983	Positive sentiment, success states
semantic.warning	#947C4A	#C1AE89	Pending, caution
semantic.error	#97524C	#C07169	Negative sentiment, errors
semantic.info	#4F78A0	#7694B2	Information, hints
border.subtle	#E5E3DC	#2A2925	Default borders, dividers
border.default	#C3BBA4	#3D3A33	Input borders
border.strong	#957F3D	#635D49	Emphasized borders

2.2 Typography Scale

Type uses a modular scale with a 1.250 ratio (major third). Three families: **Space Grotesk** for display, **Inter** for UI/body, **JetBrains Mono** for code and meta. All three are loaded as variable fonts with full weight ranges.

Token	Font	Size/Line/Weight	Track	Usage
display.xl	Space Grotesk	72 / 80 / 900	-2%	Hero headlines
display.lg	Space Grotesk	56 / 64 / 800	-1.5%	Landing hero
display.md	Space Grotesk	40 / 48 / 700	-1%	Page titles
display.sm	Space Grotesk	32 / 40 / 700	-0.5%	Section titles
heading.lg	Inter	24 / 32 / 700	-0.25%	H2 in app
heading.md	Inter	20 / 28 / 600	0	H3 in app
heading.sm	Inter	16 / 24 / 600	0	H4 in app
body.lg	Inter	18 / 28 / 400	0	Long-form reading
body.md	Inter	16 / 24 / 400	0	Default body
body.sm	Inter	14 / 20 / 400	0	UI body, table cells
caption	Inter	12 / 16 / 500	+0.5%	Captions, labels
overline	JetBrains Mono	11 / 16 / 500	+1.5%	Kickers, eyebrows (uppercase)
code.md	JetBrains Mono	14 / 20 / 400	0	Code blocks

2.3 Spacing & Layout

Spacing uses a 4px base unit. Layout uses a 12-column grid at desktop ($\geq 1024\text{px}$), 8-column at tablet ($\geq 640\text{px}$), 4-column at mobile ($< 640\text{px}$). Maximum content width for marketing pages: 1200px. For app: 1440px (dense data UIs need width).

Token	Value	Usage
space.0	0px	Touching
space.1	4px	Icon-to-text
space.2	8px	Tight groupings
space.3	12px	Related items
space.4	16px	Default padding
space.6	24px	Card padding
space.8	32px	Section gap
space.12	48px	Page section gap
space.16	64px	Hero spacing
space.24	96px	Page-level breaks
radius.sm	6px	Buttons, inputs
radius.md	8px	Cards
radius.lg	12px	Modals
radius.xl	16px	Hero cards
radius.full	9999px	Pills, avatars

3. Component Library

The component library is built on top of Radix UI primitives (for accessibility) + Tailwind (for styling) + Framer Motion (for animation). This is the same stack used by Linear, Cal.com, and shadcn/ui. The library is organized into 4 tiers.

Tier 1 — Core Primitives (25 components)

- **Button** — 5 variants (primary, secondary, ghost, destructive, outline), 4 sizes (xs, sm, md, lg), loading state, icon slot, magnetic hover effect
- **Input** — text, email, password, search, with leading/trailing icon, error state, character counter
- **Textarea** — auto-resize, character counter, mention support
- **Select** — searchable, multi-select, async (debounced search), custom render
- **Checkbox** — indeterminate state, custom icon

- **Radio** — card variant, inline variant
- **Switch** — with label, description, loading state
- **Slider** — range, multi-thumb, with tooltips
- **DatePicker** — single, range, with shortcuts
- **TimePicker** — 12/24 hour, with timezone
- **Badge** — 5 colors (default, success, warning, error, info), with dot, dismissible
- **Tag** — multi-color, with avatar, removable
- **Avatar** — with fallback initials, status dot, group stacking
- **Tooltip** — with delay, arrow, rich content
- **Popover** — with anchor, dismissable
- **Dropdown** — with search, keyboard nav
- **Modal** — 3 sizes, with header/footer, dismissable
- **Dialog** — alert, confirm, prompt variants
- **Drawer** — left, right, bottom, full screen
- **Toast** — 4 variants, with action, promise-based API
- **Skeleton** — pulse, with shape variants (text, circle, rect)
- **Progress** — linear, circular, with label
- **Spinner** — 3 sizes, with label
- **Tabs** — underline, pill, segmented variants
- **Accordion** — single, multi, with chevron

Tier 2 — Data & Loading (10 components)

- **DataTable** — sortable, filterable, paginated, with bulk select, row expansion, virtualization
- **EmptyState** — with illustration, headline, description, CTA
- **ErrorState** — with illustration, error code, retry CTA
- **Pagination** — page numbers, with jump-to
- **FilterBar** — multi-facet, with active count, clear all
- **SortMenu** — dropdown with active indicator
- **StatCard** — number, label, delta vs period, sparkline
- **Chart** — Recharts wrapper (line, bar, area, pie, radar, heatmap)
- **Sparkline** — minimal inline chart
- **ProgressRing** — circular progress with center label

Tier 3 — Patterns (8 components)

- **CommandPalette** — Cmd+K, fuzzy search, recent items, actions, navigation
- **QuickActions** — floating action button with action menu
- **MagneticButton** — cursor-tracking magnetic effect on hover
- **GradientBorder** — animated gradient border on hover

- **GlassCard** — backdrop-blur glass effect
- **AuroraBackground** — animated gradient mesh background
- **ScrollVideo** — scroll-linked video playback (for landing)
- **BentoGrid** — responsive bento-box layout

Tier 4 — Marketing (10 components)

- **AnimatedHero** — word-by-word reveal with gradient text
- **Marquee** — infinite logo carousel
- **FAQAccordion** — with search
- **PricingTable** — monthly/annual toggle, feature comparison, CTA
- **TestimonialCarousel** — with dot nav, autoplay
- **BeforeAfter** — draggable slider comparison
- **LiveDemo** — interactive product demo with typewriter
- **StatCounter** — animated count-up on scroll
- **ScrollReveal** — fade-in-up on scroll into view
- **FooterCTA** — final-conversion CTA with gradient bg

4. Motion System

Motion is built on Framer Motion. Every animation is defined as a token in `lib/design/motion.ts`, so motion is consistent across the app and tunable in one place. Three principles: (1) every state change is animated, (2) animations respect prefers-reduced-motion, (3) no animation exceeds 400ms.

Token	Properties	Duration	Easing	Usage
<code>motion.fade-in</code>	opacity 0→1, y 8→0	250ms	ease-out	Default reveal
<code>motion.fade-in-up</code>	opacity 0→1, y 16→0	350ms	ease-out	Section reveals
<code>motion.scale-in</code>	opacity 0→1, scale 0.96→1	200ms	ease-out	Modal open
<code>motion.slide-in-right</code>	x 100%→0	300ms	ease-in-out	Drawer open
<code>motion.slide-in-bottom</code>	y 100%→0	300ms	ease-in-out	Mobile sheet
<code>motion.hover-lift</code>	y 0→-2	150ms	ease-out	Card hover
<code>motion.tap-scale</code>	scale 1→0.97	100ms	ease-out	Button tap
<code>motion.spring-bounce</code>	scale, with spring	400ms	spring	Notification badge
<code>motion.shimmer</code>	background-position 0→100%	1500ms	linear	Skeleton loader
<code>motion.glow-pulse</code>	box-shadow intensity	2000ms	ease-in-out	Active states

5. Information Architecture

The app is organized into 6 top-level sections, accessible via the sidebar nav. The marketing site has 8 top-level pages. Mobile uses a bottom tab bar with 5 items.

5.1 App Navigation (Sidebar)

Route	Label	Description	Icon
/	Dashboard	Overview of metrics, recent reviews, alerts	Home icon
/inbox	Inbox	Unified review inbox with filters	Inbox icon
/reviews	Reviews	All reviews with reply management	Star icon
/requests	Requests	Review request campaigns + send history	Send icon
/analytics	Analytics	Sentiment, topics, trends, competitor	Chart icon
/widgets	Widgets	Embeddable widget builder + testimonials	Code icon
/reports	Reports	Scheduled reports + executive dashboards	File icon
/settings	Settings	Business profile, integrations, billing, team	Settings icon

5.2 Marketing Navigation

Route	Label	Description
/	Home	Hero, social proof, features, pricing, CTA
/features	Features	10 module deep-dives with interactive demos
/solutions/[slug]	Solutions	Per-vertical pages (dental, hospitality, agency, etc.)
/vs/birdeye	vs Birdeye	Comparison page with feature matrix
/vs/podium	vs Podium	Comparison page with feature matrix
/pricing	Pricing	4 tiers (Free, Starter \$49, Pro \$99, Enterprise \$299)
/about	About	Company story, team, investors, careers
/blog	Blog	SEO content + customer stories

6. Responsive Breakpoints

Mobile-first. Six breakpoints covering all common device classes. Every page is designed at mobile first, then enhanced at each larger breakpoint.

Token	Range	Device	Layout
xs	< 480px	Small mobile (iPhone SE)	Single column, bottom tab bar

Token	Range	Device	Layout
sm	480–639px	Large mobile (iPhone 14)	Single column, bottom tab bar
md	640–1023px	Tablet (iPad)	Two column where appropriate, bottom tab bar
lg	1024–1279px	Small laptop	Sidebar visible, multi-column
xl	1280–1535px	Desktop	Full sidebar, multi-column
2xl	≥ 1536px	Large desktop	Full sidebar, max content width

7. Accessibility (WCAG 2.1 AA)

- All interactive elements have visible focus rings (2px solid accent, 2px offset)
- Color contrast ≥ 4.5:1 for body text, ≥ 3:1 for large text and UI components
- All form inputs have associated elements
- All images have alt text (decorative images have alt="")
- All modals trap focus and restore on close
- Keyboard navigation works for every interaction (Tab, Shift+Tab, Enter, Space, Esc)
- Skip-to-main-content link on every page
- ARIA live regions for dynamic content (new review arrives, toast appears)
- prefers-reduced-motion respected (animations disabled or simplified)
- prefers-color-scheme respected (auto-switch to dark mode)
- Screen reader tested with VoiceOver (macOS), NVDA (Windows), TalkBack (Android)

8. Figma File Structure Recommendation

For the design team, the Figma file should be organized as follows. This structure mirrors the token system in code, enabling a 1:1 mapping between design and implementation.

- **Cover & TOC** — file metadata, version history, change log
- **01 — Foundations** — color, typography, spacing, radius, shadow, motion tokens
- **02 — Icons** — lucide icon set with custom brand additions
- **03 — Primitives** — Button, Input, Select, Modal, Toast, etc. (Tier 1 + 2)
- **04 — Patterns** — CommandPalette, QuickActions, EmptyState, etc. (Tier 3)
- **05 — Marketing** — landing page, pricing, comparison, blog templates (Tier 4)
- **06 — App Pages** — Dashboard, Inbox, Reviews, Requests, Analytics, Settings
- **07 — Email Templates** — review request, digest, win-back, onboarding
- **08 — Widget Themes** — 6 widget layouts × 4 color themes
- **09 — Mobile** — every app page designed mobile-first
- **10 — Dark Mode** — every primitive and page in dark variant

- **11 — White-Label** — agency theme variants (3 reference themes)
- **12 — Prototypes** — interactive Figma prototypes for user testing

Design Architecture — Bottom Line

The design system is the single highest-leverage investment in the rebuild. It enables every subsequent feature to ship at premium quality without bespoke design work. The 53-component library takes 3 weeks to build (Phase 1) and pays for itself across the remaining 21 weeks of feature work. Skip it and the app will look like Birdeye — functional but not premium.

Part VI — Development Architecture

SYSTEM · DATA · API · SECURITY · INFRASTRUCTURE

This part specifies the complete development architecture for ReviewReply Enterprise. It covers: system architecture, multi-tenant data model, API design, authentication & authorization, background jobs, integration layer, security architecture, observability, infrastructure, CI/CD, and disaster recovery. The goal is a system that scales from 0 to 100,000 businesses without re-architecture.

1. System Architecture Overview

The system is a serverless-first architecture on Vercel + Supabase, with edge functions for latency-sensitive paths and Inngest for background job orchestration. This stack minimizes operational overhead while scaling to 100k+ businesses.

- **Frontend:** Next.js 14 App Router on Vercel. Server Components for data-heavy pages, Client Components for interactivity. Edge runtime for auth middleware.
- **Backend:** Next.js Route Handlers (Node runtime) + Vercel Edge Functions for low-latency paths. No separate backend service — keeps the deploy surface small.
- **Database:** Supabase Postgres (managed). Row-Level Security (RLS) for multi-tenant isolation. Read replicas via Supabase connection pooler (PgBouncer) at scale.
- **Auth:** Supabase Auth (email/password, magic link, Google, Apple, SAML for enterprise). JWT in httpOnly cookies. Edge middleware for session refresh.
- **Background Jobs:** Inngest for all async work (review polling, AI draft generation, email send, scheduled reports, churn rescue).
- **Realtime:** Supabase Realtime (Postgres changes broadcast) for live inbox updates.
- **Storage:** Supabase Storage for user uploads (logos, brand voice samples). Cloudflare R2 for widget CDN.
- **CDN:** Vercel Edge Network for app + landing. Cloudflare for widget assets.
- **Email:** Resend for transactional. Customer.io for lifecycle/marketing (separate concern).
- **SMS/WhatsApp:** Twilio (SMS + WhatsApp Business).
- **Payments:** Stripe (checkout, customer portal, webhooks, metered billing).
- **AI:** Anthropic Claude 3.5 Sonnet (default), OpenAI GPT-4o (fallback), with abstraction layer for cost optimization.
- **Observability:** Sentry (errors), PostHog (product analytics), Vercel Analytics (web vitals), Datadog (infra metrics) — Datadog only at scale.
- **Customer Support:** Plain.com (in-app) + Zendesk (email) at scale.

2. Multi-Tenant Data Model

The data model uses a hierarchical multi-tenant structure: Organization → Businesses → Locations → Users. RLS policies enforce isolation at every level. The model supports both single-business accounts (a restaurant

owner) and agency accounts (an agency managing 50 client businesses).

2.1 Core Entity Hierarchy

Table	Description	Key Columns
organizations	Top-level tenant. Has plan, billing, owner.	id, name, plan, stripe_customer_id, created_at
org_members	User ↔ Org membership with role.	id, org_id, user_id, role, joined_at
businesses	A business (e.g., a restaurant). Belongs to an org.	id, org_id, name, industry, address, timezone, gb_location_id
locations	A specific location of a multi-location business.	id, business_id, name, address, phone, gb_location_id
users	A person who can log in. Has email, name, avatar.	id, email, name, avatar_url, created_at
reviews	A review from Google/FB/etc. on a business.	id, business_id, source, external_id, author, rating, text, created_at
review_threads	Groups reviews from same customer.	id, business_id, customer_hash, review_ids[]
review_requests	A request sent to a customer asking for a review.	id, business_id, customer_contact, channel, status, sent_at, clicked_at, converted_at
reply_drafts	AI-generated draft reply, pending approval.	id, review_id, draft_text, status, generated_by, generated_at
reply_approvals	Approval/rejection record for a draft.	id, draft_id, approver_id, status, decided_at, edited_text
campaigns	A multi-channel review-request campaign.	id, business_id, name, trigger_config, audience_filter, channel_mix, message_variants
campaign_events	Per-recipient event for a campaign.	id, campaign_id, customer_id, channel, status, sent_at, delivered_at, clicked_at

2.2 Enterprise Extensions

Table	Description	Key Columns
audit_logs	Every write operation, for SOC2.	id, actor_id, action, target_type, target_id, metadata, ip, created_at
webhook_events	Inbound webhook receipts for idempotency.	id, source, event_id, payload, signature, received_at, processed_at, status
oauth_tokens	Encrypted Google/FB/Apple OAuth tokens.	id, business_id, provider, access_token_enc, refresh_token_enc, expires_at, scopes

Table	Description	Key Columns
api_keys	Public API keys for customer integrations.	id, org_id, name, hash, last_used, created_at, revoked_at
sentiment_analyses	Cached AI sentiment + topic analysis per review.	id, review_id, score, topics, model, created_at
competitor_snapshots	Weekly competitor data snapshot.	id, business_id, competitor_name, rating, review_count, sentiment_topics, snapshot_at
brand_voice_profiles	Per-business trained AI voice profile.	id, business_id, voice_signature_enc, few_shot_examples, last_trained_at, accept_rate
usage_events	Metered billing events (SMS sent, emails sent, AI calls).	id, business_id, event_type, units, period, created_at
feature_flags	Per-org feature flag overrides.	id, org_id, key, enabled, rollout_pct, target_orgs
widgets	Embeddable widget configurations.	id, business_id, layout, filters, theme, embed_snippet, created_at

2.3 RLS Policy Strategy

Every table has RLS enabled. The pattern is: a user can only access rows where the owning organization has them as a member. This is enforced by a single SQL function `auth.user_orgs()` that returns the set of `org_ids` the current JWT belongs to. RLS policies then check `org_id IN auth.user_orgs()`.

Agency mode adds a second layer: an agency user can access rows where the org is a client of one of their agency orgs. This is enforced by `auth.user_agencies()` which returns the set of agency `org_ids`. The combined policy is: `org_id IN auth.user_orgs() OR org_id IN (SELECT client_org_id FROM agency_clients WHERE agency_org_id IN auth.user_agencies())`.

3. API Design

RESTful API with predictable resource naming. All responses are JSON. Errors use RFC 7807 Problem Details format. Pagination is cursor-based (not offset) for performance on large tables. All list endpoints support filtering, sorting, and field selection.

3.1 REST Conventions

- **Resource naming:** plural nouns (`/api/businesses`, `/api/businesses/[id]/reviews`)
- **HTTP methods:** GET (read), POST (create), PATCH (update), DELETE (remove), PUT (rarely, for full replaces)
- **Status codes:** 200 (OK), 201 (Created), 204 (No Content), 400 (Bad Request), 401 (Unauthorized), 403 (Forbidden), 404 (Not Found), 409 (Conflict), 422 (Unprocessable), 429 (Too Many Requests), 500 (Server Error)
- **Pagination:** cursor-based, with `cursor`, `limit` (default 50, max 100), `has_more`
- **Filtering:** query params (`?status=active&rating_gte=4`)
- **Sorting:** `?sort=-created_at,name` (minus prefix = descending)

- **Field selection:** ?fields=id,name,rating (whitelist)
- **Idempotency:** POST accepts Idempotency-Key header (stored for 24h)
- **Versioning:** URL-based (/api/v1/..., /api/v2/...)

3.2 Error Response Format (RFC 7807)

Every error response follows RFC 7807 Problem Details for HTTP APIs. This makes error handling consistent and machine-parseable. Example response shape:

```
HTTP/1.1 422 Unprocessable Entity
Content-Type: application/problem+json

{
  "type": "https://docs.reviewreply.com/errors/validation",
  "title": "Validation failed",
  "status": 422,
  "detail": "The review_id field is required.",
  "instance": "/api/v1/reviews/draft",
  "errors": [
    {"field": "review_id", "code": "required", "message": "Review ID is required"}
  ]
}
```

4. Authentication & Authorization

Supabase Auth handles all credential storage and session management. Sessions are JWT stored in httpOnly Secure cookies. Edge middleware refreshes the JWT on every request if it is within 60 seconds of expiry. Role-based access control (RBAC) is enforced at three layers: (1) RLS at the database, (2) middleware at the route, (3) UI at the component (hide/disable buttons the user cannot use).

4.1 RBAC Matrix

Role	Permissions	Scope
Owner	Full access including billing, delete org, manage members	All resources
Admin	Everything except billing and org deletion	All resources except /billing
Staff	Read/write reviews, requests, analytics. No team management.	Reviews, Requests, Analytics
Viewer	Read-only access to everything	All resources (read-only)
Agency Admin	Manages agency org + all client orgs	Agency + all client orgs
Agency Staff	Manages assigned client businesses only	Assigned client businesses
Client Admin	Full access to their own client org	Own client org only

Role	Permissions	Scope
Client Staff	Read/write their own reviews	Own client reviews only

4.2 SSO (SAML / OIDC) for Enterprise

Enterprise tier (\$299+/mo) supports SAML 2.0 and OIDC SSO. Implementation uses Supabase Auth's enterprise SSO features. Configuration flow: enterprise customer uploads SP metadata → ReviewReply provides IdP metadata → customer configures IdP → test login → roll out. Supported IdPs: Okta, Azure AD, Google Workspace, OneLogin.

5. Background Jobs (Inngest)

Inngest is the single background-job system. Every async task is an Inngest function with: trigger (event | schedule | manual), retry policy, concurrency limit, and dead-letter queue. All functions are defined in lib/inngest/ and registered via the single /api/inngest webhook (with signature verification).

5.1 Inngest Function Catalog

Function	Trigger	Description	Timeout (s)
poll-reviews	Every 15 minutes per business	Pull new reviews from Google/FB/etc., store in DB	5
generate-reply-draft	On review:created event	Generate AI draft reply for new review	3
send-review-request	On campaign:send event	Send SMS/email/WhatsApp per recipient	50
send-batch-review-requests	On campaign:send-batch event	Batch send with rate limiting	10
post-reply	On reply:approved event	Post approved reply to Google/FB	3
daily-digest	Cron: 9am user-local	Generate and send daily digest email	60
weekly-summary	Cron: Monday 9am user-local	Generate and send weekly summary PDF	60
monthly-executive	Cron: 1st of month 9am user-local	Generate and send monthly executive PDF	120
trial-expiration-warning	Cron: daily 12pm UTC	Send trial-expiring-soon email	60
pii-retention-purge	Cron: daily 2am UTC	Delete customer_contact older than 90 days	120
oauth-token-refresh	Cron: hourly	Refresh Google/FB tokens expiring within 1 hour	60

Function	Trigger	Description	Timeout (s)
competitor-snapshot	Cron: weekly Monday 6am UTC	Pull competitor data, store snapshot	300
stripe-subscription-sync	Cron: every 6 hours	Reconcile local subscription state with Stripe	60
ai-voice-retrain	On brand_voice:retrain event	Re-train per-business AI voice profile	300
sentiment-anomaly-detect	Cron: daily 4am UTC	Detect sentiment anomalies, create alerts	120
abandoned-cart-recovery	On checkout:abandoned event (3 day delay)	Send nudge email to abandoned checkout	60

6. Security Architecture

Security is layered: network (Cloudflare WAF + DDoS), edge (Vercel Edge rate limiting), application (Zod input validation, CSRF tokens, RBAC), database (RLS, encrypted columns), and operational (audit logs, secret rotation, pen tests).

6.1 PII Encryption

Customer contact information (phone, email) is encrypted at rest using AES-256-GCM with a KMS-managed key (Supabase Vault). Encryption happens in the application layer before insert, so even DBAs with direct DB access cannot read PII. Decryption happens in the application layer on read, with audit logging of every decryption event.

6.2 Rate Limiting

Endpoint	Limit	Purpose
/api/auth/*	5 per minute per IP	Brute-force protection
/api/businesses/*/review-requests/send-batch	10 per hour per org	SMS cost abuse
/api/reviews/[id]/draft	20 per minute per org	AI cost abuse
/api/ingest	100 per second	Inngest webhook flood
/api/webhooks/stripe	50 per second	Stripe webhook flood
All other endpoints	100 per minute per user	General abuse

6.3 Audit Logs

Every write operation (create, update, delete) is logged in the audit_logs table with: actor_id, action, target_type, target_id, metadata (JSON of changed fields), IP address, user agent, and timestamp. Logs are

retained for 7 years for SOC2 and GDPR compliance. A dedicated audit log UI in Settings lets org owners search and export logs.

7. Observability

- **Sentry**: error tracking + performance monitoring. Source maps uploaded on every deploy. Alerting into Slack #engineering-alerts.
- **PostHog**: product analytics (funnels, retention, feature flags, session replays). Auto-capture on, with PII filtering.
- **Vercel Analytics**: Core Web Vitals + audience insights.
- **Datadog**: at scale (>\$50k MRR) — infra metrics, APM, log aggregation, synthetic monitoring.
- **Logflare / Supabase Logs**: application logs, structured JSON, 30-day retention.
- **Status page**: status.reviewreply.com (powered by Atlassian Statuspage or Better Stack).
- **Uptime monitoring**: Better Stack or Pingdom, 1-minute checks on /api/health, alert if 2 consecutive failures.

8. Infrastructure & CI/CD

The infra is intentionally serverless-first to minimize operational overhead. CI/CD is GitHub Actions → Vercel preview deploy per PR → automated tests → merge to main → production deploy. Database migrations run via Supabase CLI as part of the deploy pipeline.

- **Frontend deploy**: Vercel, auto-deploy on main merge. Preview deploys per PR.
- **Database migrations**: Supabase CLI, applied via GitHub Actions on merge to main. Roll-forward only — no destructive migrations without a paired forward migration.
- **Edge functions**: deployed via Vercel CLI as part of frontend deploy.
- **Inngest functions**: deployed via Inngest CLI on merge to main.
- **Widget CDN**: Cloudflare R2 + Workers, deployed via wrangler CLI.
- **Secrets**: Vercel Environment Variables (encrypted at rest), Doppler for local dev.
- **Feature flags**: PostHog feature flags, with database override for per-org targeting.
- **Staging environment**: full staging stack on Vercel + Supabase staging project. E2E tests run against staging before production deploy.

9. Disaster Recovery

Component	Backup Cadence	RTO	Restore Method
Database (Supabase)	Daily automated + 6-hour PITR	15 minutes	Supabase dashboard + CLI restore
Vercel deploys	Every deploy is rollback-able	< 1 minute	Vercel dashboard one-click rollback

Component	Backup Cadence	RTO	Restore Method
Inngest functions	Function code in git, deployable from any commit	< 5 minutes	Re-run Inngest CLI deploy
Widget CDN (R2)	Source in git, redeployable	< 5 minutes	Re-run wrangler deploy
Secrets (Doppler)	Versioned, restorable to any prior version	< 1 minute	Doppler dashboard
Audit logs	Replicated to S3 (read replica)	< 1 hour	Manual restore from S3

RPO (Recovery Point Objective): 15 minutes (PITR). **RTO (Recovery Time Objective):** 1 hour for full system restoration. A quarterly disaster-recovery drill is mandatory: restore staging DB to a fresh Supabase project, verify all critical user journeys work, document any gaps, file issues for fixes.

Development Architecture — Bottom Line

The serverless-first stack (Vercel + Supabase + Inngest) is correct for ReviewReply's scale (0–100k businesses). It minimizes operational overhead, scales automatically, and lets a small team ship fast. The architecture's main risk is vendor lock-in — mitigated by keeping business logic in framework-agnostic modules under lib/ so a future migration to AWS/Railway/Render is possible without rewriting the app.

Part VII — Phased Implementation Roadmap

24 WEEKS · 7 PHASES · RACI · EXIT CRITERIA · KPIS

This part sequences the work into seven phases over twenty-four weeks. Each phase has clear deliverables, dependencies, an exit criteria checklist, and KPIs. The phases are designed to deliver visible value every two weeks — no phase ends without something a user or stakeholder can touch.

Phase 0 — Stabilization (Week 1–2, \$15k)

FIX ALL P0/P1 BUGS · ISOLATE MOCK MODE · PATCH SECURITY HOLES

Before adding anything new, fix every P0 and P1 bug identified in Part II. This phase is non-negotiable — no new feature work begins until the existing app is correct and secure. Mock mode is clearly isolated behind a feature flag, security holes are patched, and the test suite is brought up to a baseline coverage.

Deliverables

- Fix all 5 P0 bugs (FB-01 through FB-05, BB-01 through BB-05)
- Fix all 14 P1 bugs (FB-06 through FB-10, BB-07 through BB-10, plus design P1s)
- Add `isMockMode()` guards to every integration file — mock data must never leak into production paths
- Add `app/not-found.tsx`, `app/error.tsx`, `app/loading.tsx`
- Add `app/(marketing)/layout.tsx` with shared nav + footer
- Fix invalid Tailwind classes (`space-y-22`, `mb-22`, `duration-400`, `prose prose-invert`)
- Add Stripe webhook handler at `app/api/webhooks/stripe/route.ts` with signature verification
- Add Inngest signature verification to `app/api/inngest/route.ts`
- Add `/api/dev/trigger` authentication (admin-only)
- Fix `AI_MODEL` to `claude-3-5-sonnet-20241022`
- Add `subscription_plan` enum value `enterprise` via migration 0003
- Add RLS policies to organizations table
- Add rate limiting middleware (Upstash Redis)
- Baseline test coverage: unit tests for `lib/ai`, `lib/parsing`, `lib/integrations`

Exit Criteria

- All P0/P1 bugs closed in tracker
- Mock mode clearly isolated — `grep -r 'mock-data' app/` returns zero hits in production paths
- Stripe webhook test event processed successfully
- Inngest signed event accepted; unsigned event rejected with 401
- Test coverage $\geq 60\%$ on `lib/`
- Security scan (Snyk) shows zero high-severity vulnerabilities

KPIs

- P0/P1 bug count: 23 → 0

- Mock data leak count: many → 0
- Test coverage: ~0% → 60%+
- Security vulnerabilities (high+): many → 0

Phase 1 — Design System (Week 3–5, \$45k)

COMPONENT LIBRARY · MOTION · DARK MODE · MOBILE-FIRST

Build the 53-component library defined in Part V §3. Install Radix UI + Tailwind + Framer Motion + lucide-react. Build the token system. Replace every emoji icon with lucide. Build dark mode. Build mobile-first patterns. By end of Phase 1, the existing app pages should look 80% premium — they will use the new components without yet adding new features.

Deliverables

- Install: radix-ui, framer-motion, lucide-react, recharts, cmdk, react-hook-form, zod
- Token system: lib/design/tokens.ts with color, typography, spacing, radius, motion
- Tier 1 primitives (25 components): Button, Input, Modal, Toast, Tabs, etc.
- Tier 2 data components (10): DataTable, Skeleton, EmptyState, StatCard, Chart
- Tier 3 patterns (8): CommandPalette, MagneticButton, GlassCard, AuroraBackground
- Dark mode: every component in light + dark, with prefers-color-scheme auto-switch
- Replace every emoji icon with lucide-react equivalent
- Replace every Unicode icon (★ ☆ ✓ → ● ◆) with lucide
- Build app/(marketing)/layout.tsx with shared responsive nav + footer + mobile hamburger
- Build mobile bottom tab bar for app
- Add skeleton loaders to every async page
- Add toast notifications for every user action (save, error, success)
- Add Framer Motion page transitions (route-level)

Exit Criteria

- All 53 components built, documented in Storybook
- Every existing page uses new components (no raw Tailwind for primitives)
- Dark mode toggles correctly on every page
- Mobile responsive at 360px, 414px, 768px, 1024px, 1440px, 1920px
- Lighthouse mobile performance ≥ 90 on top 5 pages
- WCAG 2.1 AA audit passes (axe DevTools)

Phase 2 — Auth & Multi-Tenant (Week 6–7, \$30k)

SUPABASE AUTH · RBAC · SSO · ORGANIZATIONS · AUDIT LOGS

Wire real Supabase Auth (currently broken). Build the RBAC system from Part VI §4. Add SSO for enterprise tier. Build the organizations + members UI. Add audit logs.

Deliverables

- Real Supabase Auth: signup, login, password reset, email verification, magic link
- OAuth providers: Google, Apple (consumer); SAML + OIDC (enterprise)
- RBAC: 8 roles per Part VI §4.1, enforced at RLS + middleware + UI
- Organizations UI: create, invite members, manage roles
- Audit log table + UI in Settings (searchable, exportable)
- API keys: generate, rotate, revoke (for public API access)
- Session management: edge middleware refresh, httpOnly cookies, secure flag
- Password policy: ≥ 12 chars, ≥ 1 number, ≥ 1 symbol, breach check (HaveIBeenPwned API)
- 2FA: TOTP via authenticator app, backup codes

Exit Criteria

- User can sign up, verify email, log in, reset password without mock mode
- Google OAuth login works end-to-end
- SAML SSO works with Okta test tenant
- RBAC: every role sees only what they should (automated test suite)
- Audit log records every write operation
- 2FA enrollment + login + backup codes work

Phase 3 — Core Module Rebuild (Week 8–11, \$60k)

INBOX · REQUESTS · AI REPLY · BILLING WEBHOOKS · STRIPE CUSTOMER PORTAL

Rebuild the four core operational modules. This is where the existing scaffold gets upgraded to production-grade. Module 1 (Unified Inbox), Module 2 (Multi-Channel Requests), Module 3 (AI Reply with Brand Voice), and the billing webhooks / customer portal.

Deliverables

- Module 1 — Unified Inbox (2 weeks)
- Module 2 — Multi-Channel Request Automation (2 weeks, without WhatsApp — added in Phase 5)
- Module 3 — AI Reply with Brand Voice Training (3 weeks, overlaps with Phase 4 start)
- Stripe billing: webhook handler, customer portal, invoice sync, metered billing
- Stripe customer portal embedded at /billing/portal
- Real Google Business Profile OAuth: fetchReviews + postReply working
- Real Facebook Graph OAuth: fetchReviews + postReply working
- Twilio 10DLC registration flow + opt-out handling
- Resend bounce/complaint webhook handling
- Click tracking: app/r/[token]/page.tsx live

Exit Criteria

- End-to-end: business signs up → connects Google → review arrives in inbox → AI draft generated → approved → posted to Google
- End-to-end: business creates campaign → sends 100 SMS → 30 click through → 8 leave review → 6 are positive
- Stripe: checkout → webhook → subscription active → customer portal accessible → cancel takes effect at period end
- Brand voice: trained on 50+ historical replies, drafts sound like the business

Phase 4 — New Enterprise Modules (Week 12–18, \$105k)

AGENCY · SENTIMENT · WIDGETS · COMPETITOR · REPORTING

Build the five enterprise modules that expand TAM and defensibility. These run in parallel where possible — Modules 4, 5, 7, 8 can be built by separate engineers simultaneously. Module 6 (Agency) is the most complex and gets a dedicated engineer.

Deliverables

- Module 4 — Sentiment & Topic Analytics (2 weeks)
- Module 5 — Widget & Testimonial Engine (2 weeks)
- Module 6 — Multi-Tenant Agency Mode (2 weeks)
- Module 7 — Competitor Intelligence (2 weeks)
- Module 8 — Reporting & Alerts (1.5 weeks)
- White-label: custom domain, logo, colors, CSS override
- Slack + Microsoft Teams integrations (OAuth + alert routing)
- Scheduled PDF reports (daily, weekly, monthly)
- Executive dashboard + agency dashboard

Phase 5 — Automation & AI (Week 19–22, \$60k)

BRAND VOICE · MULTI-LANGUAGE · SMART ROUTING · CHURN RESCUE · WHATSAPP

Deepen the AI and automation layer. Brand voice fine-tuning loop. Multi-language support. Smart send-time optimization. Churn rescue flows. WhatsApp + Apple Business Chat channels.

Deliverables

- Brand voice feedback loop: track accept/reject/edit rate per business, re-train weekly
- Multi-language: detect review language, draft reply in same language (26 languages)
- Smart send-time: ML model for optimal send hour per recipient
- A/B testing: variant selection, statistical significance, auto-promotion
- Churn rescue: detect at-risk customers (sentiment drop, NPS detractor), trigger outreach workflow
- WhatsApp Business: template approval, opt-in flow, send + receive
- Apple Business Chat: invitation flow, message templates
- Detractor rescue flow: NPS 0–6 triggers automatic outreach

- Programmatic local SEO: JSON-LD schema generator, embed snippet, velocity tracker

Phase 6 — Compliance & Launch (Week 23–24, \$30k)

SOC2 · GDPR · TCPA · PERFORMANCE · BETA · PUBLIC LAUNCH

Final hardening for commercial launch. SOC2 Type I readiness (Type II starts at launch). GDPR/CCPA/TCPA compliance verification. Performance optimization. Closed beta with 20 design partners. Public launch at end of Week 24.

Deliverables

- SOC2 Type I readiness: policies, controls, evidence collection, auditor engagement
- GDPR: DPA template, data subject access request (DSAR) flow, right-to-erasure automation
- CCPA: 'Do Not Sell My Info' page, opt-out flow
- TCPA: opt-in checkbox, consent timestamp, 10DLC campaign registration, quiet hours
- Performance: Core Web Vitals LCP < 2.5s, FID < 100ms, CLS < 0.1 on all top pages
- Accessibility: WCAG 2.1 AA audit passes (third-party audit)
- Penetration test: third-party pen test, fix all criticals
- Closed beta: 20 design partners, weekly feedback synthesis, iterate
- Public launch: landing page live, Product Hunt launch, Hacker News Show HN
- Status page (status.reviewreply.com), SLA template, on-call rotation

Launch Go/No-Go Checklist

- All P0/P1/P2 bugs closed
- Pen test: zero criticals, zero highs
- SOC2 Type I report received
- GDPR DSAR flow tested with real data
- Core Web Vitals: green on all top pages
- WCAG 2.1 AA: third-party audit passes
- Backup restore drill successful
- On-call rotation documented
- Status page live
- Customer support runbook documented
- Pricing page live
- Billing: test end-to-end with real credit card
- 20 design partners actively using product
- NPS from beta ≥ 30

RACI Summary

RACI = Responsible (does the work) · Accountable (signs off) · Consulted (input) · Informed (kept in loop).
The team composition: 1 Lead Frontend Engineer (LF), 1 Lead Backend Engineer (LB), 1 Full-Stack Engineer (FS), 1 Designer (D), fractional Product Manager (PM), fractional Security Engineer (SE), fractional DevOps (DO).

Phase	Responsible	Accountable	Consulted	Informed
Phase 0 — Stabilization	LF, LB, FS	PM	SE	D
Phase 1 — Design System	D, LF, FS	PM	LB	SE
Phase 2 — Auth & Multi-Tenant	LB, FS	PM	SE, LF	D
Phase 3 — Core Module Rebuild	LF, LB, FS, D	PM	SE	—
Phase 4 — Enterprise Modules	LF, LB, FS, D	PM	—	—
Phase 5 — Automation & AI	LB, FS, LF	PM	D	—
Phase 6 — Compliance & Launch	SE, DO, PM, D	PM	LF, LB	—

Part VIII — End-to-End Automation & Manual Flows

10 CRITICAL WORKFLOWS · TRIGGERS · STEPS · SLAS · EXCEPTIONS

This part specifies ten critical end-to-end workflows. Each is defined for both the automated path (no human in the loop) and the manual path (human override). SLAs are specified per step. These flows are the contract between product, engineering, and support — when something breaks, the team knows which flow, which step, which SLA was violated.

Flow 1 — New Customer Onboarding

TRIGGER: USER SIGNS UP · SLA: 5 MIN TO FIRST VALUE

Automated Path

- Step 1: User submits signup form → Supabase Auth creates user, sends verification email (SLA: 5s)
- Step 2: User clicks verification link → marked email_verified, redirected to /onboarding (SLA: 2s)
- Step 3: User completes onboarding form (business name, industry, address, phone) → business row created, Google Business Profile auto-discovered by name+address (SLA: 10s)
- Step 4: User connects Google Business Profile (OAuth) → GBP location ID stored, initial review pull triggered (SLA: 30s)
- Step 5: First 10 reviews pulled → AI draft generated for each pending → dashboard shows '10 reviews, 7 drafts ready' (SLA: 60s)
- Step 6: Welcome email sent (Resend) with getting-started guide (SLA: 30s)

Manual Path

- If Google auto-discovery fails: user manually enters GBP URL
- If OAuth fails: user can skip and connect later, dashboard shows 'Connect Google to start receiving reviews' banner
- If no reviews found: empty state with 'Send your first review request' CTA

Flow 2 — Review Request Automation

TRIGGER: EVENT (E.G., APPOINTMENT COMPLETED) · SLA: <5 MIN FROM EVENT TO SEND

Automated Path

- Step 1: External system (POS, booking, EHR) fires webhook to /api/events with event_type=appointment_completed + customer_contact (SLA: 1s)
- Step 2: Webhook handler validates signature, dedupes by event_id (idempotency), creates review_request row (SLA: 1s)
- Step 3: Inngest function send-review-request triggered, checks send window (9am–8pm customer local), schedules send (SLA: 1s)
- Step 4: At scheduled time, SMS sent via Twilio with branded short link (SLA: 5s)

- Step 5: Customer clicks link → /r/[token] page records click, redirects to Google review page (SLA: 1s)
- Step 6: Customer leaves review → next Google poll detects it, links to review_request, marks as converted (SLA: 15 min)
- Step 7: Conversion tracked in campaign_events, dashboard updates (SLA: 1s)

Manual Path

- User can manually add a customer to a campaign via Quick Add (no webhook needed)
- User can upload a CSV of customers for bulk send (rate-limited to 100/hour)
- User can paste a list of customers (paste-list-parser handles any format)

Flow 3 — Review Polling & Ingestion

TRIGGER: CRON EVERY 15 MIN PER BUSINESS · SLA: <5 MIN FROM REVIEW-POSTED TO INBOX

Automated Path

- Step 1: Inngest poll-reviews function runs every 15 minutes per business (SLA: 1s start)
- Step 2: For each connected source (Google, FB, Yelp), call fetchReviews(since=last_poll_timestamp) (SLA: 10s)
- Step 3: New reviews inserted into reviews table, source-specific dedup by external_id (SLA: 1s)
- Step 4: review:created event fired → triggers generate-reply-draft Inngest function (SLA: 1s)
- Step 5: AI draft generated using brand voice profile (SLA: 5s)
- Step 6: Sentiment + topic analysis queued (separate Inngest function) (SLA: 5s)
- Step 7: If rating ≤ 2: real-time alert fired (Slack/email/Teams) (SLA: 5s)
- Step 8: Dashboard updates in real-time via Supabase Realtime (SLA: 1s)

Manual Path

- User can manually trigger a poll via /settings/integrations → 'Refresh now'
- If a known review is missing, user can paste the review URL for manual import

Flow 4 — AI Draft Reply Generation

TRIGGER: REVIEW:CREATED EVENT · SLA: <10S FROM REVIEW TO DRAFT

Automated Path

- Step 1: review:created event received by generate-reply-draft Inngest function (SLA: 1s)
- Step 2: Load business's brand_voice_profile (encrypted, decrypted in-memory) (SLA: 1s)
- Step 3: Construct Claude prompt: system (brand voice + rules) + 5-shot examples + user (review text) (SLA: 1s)
- Step 4: Call Anthropic API with claude-3-5-sonnet-20241022, max_tokens=300 (SLA: 5s)
- Step 5: Validate response: forbidden phrases check, escalation keyword check, length check (SLA: 1s)
- Step 6: If escalation keywords detected (lawsuit, BBB, lawyer): mark as escalated, do not auto-post, alert owner (SLA: 1s)

- Step 7: Insert reply_draft row with status=pending_approval (SLA: 1s)
- Step 8: If business has auto-post enabled AND rating ≥ 4 AND no escalation: schedule auto-post after 24h if not approved (SLA: 1s)

Manual Path

- User edits draft inline → edited text saved → brand voice feedback loop records the edit
- User rejects draft → 'Generate alternative' button creates new draft with adjusted prompt
- User can write reply from scratch (no AI) → saved as draft with generated_by=human

Flow 5 — Approval Workflow & Multi-Channel Reply Posting

TRIGGER: USER APPROVES DRAFT · SLA: <30S FROM APPROVAL TO POSTED

Automated Path

- Step 1: User clicks 'Approve & Post' on draft → reply_approvals row created (SLA: 1s)
- Step 2: post-reply Inngest function triggered (SLA: 1s)
- Step 3: Load oauth_token for source (Google/FB/etc.), refresh if expired (SLA: 2s)
- Step 4: Call source API to post reply (SLA: 5s for Google, 3s for Facebook)
- Step 5: Verify reply posted by re-fetching review (SLA: 5s)
- Step 6: Update review row with reply_text + replied_at + replied_by (SLA: 1s)
- Step 7: Toast notification: 'Reply posted to Google' (SLA: 1s)
- Step 8: If post fails (token expired, API down): retry 3x with exponential backoff, then alert user (SLA: 10s)

Manual Path

- User can copy draft text and post manually to Google/FB if API post fails persistently
- User can schedule approval for a future time (e.g., 'approve and post tomorrow at 9am')

Flow 6 — Daily Digest Email

TRIGGER: CRON 9AM USER-LOCAL · SLA: <5 MIN GENERATION

- Step 1: daily-digest Inngest function triggered per user at 9am local (SLA: 1s)
- Step 2: Query reviews received in last 24h for user's businesses (SLA: 2s)
- Step 3: Query drafts pending approval (SLA: 1s)
- Step 4: Query sentiment anomalies (SLA: 1s)
- Step 5: Generate HTML email via React Email template (SLA: 5s)
- Step 6: Send via Resend (SLA: 5s)
- Step 7: If no reviews + no pending drafts: skip send (don't spam)

Flow 7 — Real-time Negative Review Alert

TRIGGER: REVIEW:CREATED WITH RATING ≤ 2 · SLA: <30S FROM REVIEW TO SLACK

Automated Path

- Step 1: poll-reviews detects new review with rating ≤ 2 (SLA: 1s)
- Step 2: alert-negative-review Inngest function triggered (SLA: 1s)
- Step 3: Load business's alert config (Slack webhook, Teams webhook, email recipients) (SLA: 1s)
- Step 4: Construct alert message: business name, rating, review snippet, link to inbox (SLA: 1s)
- Step 5: Send to Slack (incoming webhook) + Teams (Power Automate) + email (Resend) in parallel (SLA: 5s)
- Step 6: Record alert_sent_at on review row (SLA: 1s)

Flow 8 — Weekly Competitor Benchmarking

TRIGGER: CRON MONDAY 6AM UTC · SLA: <10 MIN TOTAL

Automated Path

- Step 1: competitor-snapshot Inngest function triggered per business with competitors configured (SLA: 1s)
- Step 2: For each competitor: call Google Maps API to fetch rating + review_count (SLA: 5s per competitor)
- Step 3: Scrape last 10 reviews for sentiment topic analysis (compliant with Google TOS) (SLA: 10s per competitor)
- Step 4: Insert competitor_snapshot row (SLA: 1s)
- Step 5: Compare to last week's snapshot, compute deltas (SLA: 1s)
- Step 6: If competitor rating surpassed business: alert (SLA: 5s)
- Step 7: If competitor review velocity spiked >50% wow: alert 'Competitor likely running a campaign' (SLA: 5s)
- Step 8: Update competitor benchmark dashboard (SLA: 1s)

Flow 9 — Churn Rescue

TRIGGER: CUSTOMER MARKED AS CHURN-RISK · SLA: <1 HR FROM DETECTION TO OUTREACH

Automated Path

- Step 1: Customer Health Score (CHS) computed nightly: usage (40%), NPS (20%), support tickets (20%), sentiment (10%), billing (10%) (SLA: 60s)
- Step 2: If CHS < 50: customer marked as churn-risk (SLA: 1s)
- Step 3: churn-rescue Inngest function triggered (SLA: 1s)
- Step 4: Generate personalized outreach email (Claude with customer's usage history) (SLA: 10s)
- Step 5: Send to customer success lead for review/edit (SLA: 1s)
- Step 6: CSL reviews, edits if needed, sends (SLA: <1 hr human SLA)
- Step 7: Track response, schedule follow-up if no reply in 3 days

Flow 10 — Trial Expiration & Conversion

TRIGGER: CRON DAILY 12PM UTC · SLA: <5 MIN

Automated Path

- Step 1: trial-expiration-warning Inngest function runs daily (SLA: 1s)
- Step 2: Query businesses with trial_expires_at within next 3 days (SLA: 2s)
- Step 3: For each: send 'Your trial ends in N days' email (Resend) with upgrade CTA (SLA: 5s)
- Step 4: If trial_expires_at < 24h: also send SMS (Twilio) (SLA: 5s)
- Step 5: If trial expired: downgrade to Free tier, send 'Your trial has ended' email, retain data for 30 days (SLA: 1s)
- Step 6: If user upgrades within 30 days: restore to paid tier, all data preserved (SLA: 1s)
- Step 7: If user does not upgrade within 30 days: schedule data deletion (with 7-day grace + warning email) (SLA: 1s)

Manual Path

- Sales team can extend trial manually (up to 14 days) via admin UI
- Sales team can convert trial to paid manually with discount code

Part IX — Commercial Launch Readiness

COMPLIANCE · SECURITY · PERFORMANCE · ANALYTICS · LAUNCH CHECKLIST

This part is the gate between 'feature complete' and 'commercially launched'. It covers compliance posture (GDPR, CCPA, TCPA, HIPAA-ready, SOC2), security hardening, performance budgets, analytics instrumentation, billing & monetization tactics, support model, SLA template, and the 50-item launch go/no-go checklist.

1. Compliance Posture

1.1 GDPR (European Union)

- Data Processing Addendum (DPA) template ready for enterprise customers
- Data Subject Access Request (DSAR) flow: user requests data export → automated within 30 days
- Right to erasure: user requests deletion → all PII purged within 30 days, audit log retained (legal basis)
- Consent management: cookie consent banner, granular consent for marketing emails
- Sub-processor list published at /sub-processors, updated 30 days before adding new sub-processor
- EU data residency option (Supabase EU region) for enterprise customers
- Data Protection Officer (DPO) designated: dpo@reviewreply.com

1.2 CCPA (California)

- 'Do Not Sell My Personal Information' link in footer
- Opt-out flow: user clicks → form submission → opt-out flag set → no data shared with third parties
- Privacy policy updated to disclose categories of data collected and shared

1.3 TCPA (Telephone Consumer Protection Act)

- Explicit opt-in checkbox for SMS on every form that captures phone numbers
- Consent timestamp and IP recorded
- Quiet hours: 9pm–8am recipient local time, enforced in send logic
- 10DLC campaign registration with Twilio (required for A2P 10DLC messaging)
- Opt-out keywords: STOP, UNSUBSCRIBE, CANCEL — handled within 24 hours
- Opt-in keywords: START, YES, UNSTOP — re-enables messaging
- Message templates approved by Twilio before send

1.4 HIPAA-Ready (for healthcare vertical)

ReviewReply does not store Protected Health Information (PHI) directly, but healthcare customers may include appointment context that touches PHI. The 'HIPAA-ready' posture includes: Business Associate Agreement (BAA) available, audit logging enhanced, encryption at rest and in transit, access controls, and a designated security officer. Full HIPAA compliance requires a third-party audit and is recommended only after SOC2 Type II is achieved.

1.5 SOC2 Type II

- Phase 6 (Week 23–24): SOC2 Type I readiness — policies, controls, evidence collection
- Engage auditor (Vanta + a CPA firm like Sensiba or Prescient) for Type I
- Type I report received pre-launch (attests to control design)
- Type II starts at launch (attests to control operating effectiveness over 6–12 months)
- Continuous compliance via Vanta: automated evidence collection, real-time control monitoring

2. Security Hardening Checklist

- All API endpoints require authentication (except /api/health, /api/ready)
- All write operations require CSRF token (double-submit cookie pattern)
- All inputs validated with Zod schemas (zero any types in API routes)
- All secrets in Vercel Environment Variables, rotated quarterly
- Stripe webhook signature verification
- Inngest webhook signature verification
- Twilio webhook signature verification
- Resend webhook signature verification
- Rate limiting on all auth + AI endpoints (Upstash Redis)
- PII encrypted at rest (AES-256-GCM with KMS-managed key)
- Audit logs on every write, retained 7 years
- Penetration test by third party (Bugcrowd or Cobalt), all criticals fixed
- Bug bounty program (Bugcrowd) post-launch
- Security.txt file at /.well-known/security.txt
- Responsible disclosure policy at /security

3. Performance Budgets

Metric	Target	Note
LCP (Largest Contentful Paint)	< 2.5s	95th percentile on 4G
FID (First Input Delay)	< 100ms	95th percentile
CLS (Cumulative Layout Shift)	< 0.1	All pages
TTFB (Time to First Byte)	< 600ms	95th percentile
INP (Interaction to Next Paint)	< 200ms	95th percentile
Bundle size (initial JS)	< 200KB gzipped	Landing page
Bundle size (initial JS)	< 350KB gzipped	App dashboard

Metric	Target	Note
Image size (hero)	< 100KB	WebP, responsive srcset
API response (p50)	< 200ms	All read endpoints
API response (p95)	< 800ms	All read endpoints
API response (p99)	< 2s	All read endpoints
Inngest function start	< 5s	95th percentile

4. Analytics Instrumentation

- **PostHog**: auto-capture on (with PII filtering), funnel tracking, retention cohorts, feature flags, session replays (sampled 1%)
- **Key funnels**: signup → onboarding complete → first review request sent → first review received → first reply posted → paid conversion
- **Key events**: signup, login, business_created, google_connected, review_received, draft_generated, draft_approved, reply_posted, campaign_sent, billing_upgraded, billing_cancelled
- **Retention cohorts**: by signup week, by industry, by plan
- **NPS survey**: triggered at 30 days post-signup, in-app widget
- **Customer Health Score**: usage (40%) + NPS (20%) + support tickets (20%) + sentiment (10%) + billing (10%)
- **Vercel Analytics**: Core Web Vitals, audience insights
- **Sentry Performance**: API response times, DB query times, Inngest function durations

5. Billing & Monetization

5.1 Pricing Tiers

Tier	Price	Limits & Features	Target
Free	\$0	1 business, 50 reviews/mo, manual reply, no AI, no widgets	Acquisition
Starter	\$49/mo	1 business, 500 reviews/mo, AI draft, basic analytics, 1 widget	SMB
Pro	\$99/mo	3 businesses, unlimited reviews, brand voice, all widgets, competitor intel	Multi-location SMB
Enterprise	\$299/mo	Unlimited businesses, agency mode, SSO, white-label, priority support	Agency + multi-location
Custom	Quoted	On-prem, custom integrations, dedicated CSM, SLA	Large enterprise

5.2 Annual Pricing

Annual billing offers a 20% discount (effectively 2.4 months free). This improves cash flow and reduces churn. Pricing toggle on /pricing defaults to monthly, with a visible 'Save 20% with annual' badge on the annual toggle.

5.3 Metered Billing

SMS and AI calls are metered: each plan includes a monthly quota (Starter: 100 SMS, Pro: 500 SMS, Enterprise: 2000 SMS). Overage billed at \$0.035/SMS and \$0.02/AI call. Metered usage tracked in usage_events table, billed monthly via Stripe metered billing.

5.4 Trial Flow

- 14-day free trial of Pro tier, no credit card required
- Day 1: welcome email + getting-started checklist
- Day 3: tips email (how to get first review)
- Day 7: case study email (customer success story)
- Day 11: '3 days left' email with upgrade CTA
- Day 13: '1 day left' SMS + email
- Day 14: trial ends → downgrade to Free (data retained 30 days)
- Day 30: data purged if no upgrade (with 7-day warning)

6. Support Model

Tier	Channel	Response SLA	Uptime SLA
Free	Community forum + docs	Best-effort	None
Starter	Email (support@reviewreply.com)	24 business hours	None
Pro	Email + in-app chat (Plain.com)	4 business hours	99.5% uptime
Enterprise	Email + chat + Slack Connect	1 business hour	99.9% uptime + SLA
Custom	Dedicated CSM + phone	15 minutes	99.99% uptime + custom SLA

7. SLA Template (Enterprise)

The Enterprise SLA template includes: 99.9% uptime commitment (excluding scheduled maintenance), service credits for downtime (10% for <99.9%, 25% for <99.0%, 50% for <95.0%), 1-hour critical incident response, 4-hour high-priority response, incident postmortem within 5 business days, and quarterly business reviews.

8. Launch Go/No-Go Checklist (50 items)

All 50 items must be ✓ before public launch. Owner of each item is in brackets.

Item 1	Item 2	Item 3
☐ All P0 bugs closed [LB]	☐ All P1 bugs closed [LB]	☐ All P2 bugs closed [LB]
☐ Pen test: zero criticals [SE]	☐ Pen test: zero highs [SE]	☐ SOC2 Type I report received [SE]
☐ GDPR DSAR flow tested [LB]	☐ CCPA opt-out flow tested [LB]	☐ TCPA 10DLC registered [LB]
☐ Core Web Vitals: green on top 5 pages [LF]	☐ WCAG 2.1 AA audit passes [D]	☐ Lighthouse mobile ≥ 90 [LF]
☐ Stripe checkout end-to-end tested [LB]	☐ Stripe customer portal accessible [LB]	☐ Stripe webhook processes all event types [LB]
☐ Google OAuth end-to-end tested [LB]	☐ Facebook OAuth end-to-end tested [LB]	☐ Twilio SMS send + receive tested [LB]
☐ Resend email send + bounce handling tested [LB]	☐ Inngest all 16 functions deployed [LB]	☐ Inngest signature verification on [LB]
☐ Rate limiting on all sensitive endpoints [LB]	☐ Audit log recording every write [LB]	☐ PII encryption verified [SE]
☐ Backup restore drill successful [DO]	☐ Status page live at status.reviewreply.com [DO]	☐ On-call rotation documented [DO]
☐ Customer support runbook documented [PM]	☐ Pricing page live at /pricing [PM]	☐ Landing page live at / [PM]
☐ Blog launched at /blog [PM]	☐ Documentation live at docs.reviewreply.com [LB]	☐ API reference (OpenAPI) published [LB]
☐ 20 design partners actively using product [PM]	☐ NPS from beta ≥ 30 [PM]	☐ Case study from 3 beta customers [PM]
☐ Product Hunt launch prepared [PM]	☐ Hacker News Show HN prepared [PM]	☐ Press release drafted [PM]
☐ Sales deck ready [PM]	☐ Pricing & packaging deck ready [PM]	☐ Competitive battle cards ready [PM]
☐ Founders' social posts scheduled [PM]	☐ Email announcement to waitlist scheduled [PM]	☐ In-app announcement banner scheduled [LF]
☐ Analytics events firing correctly [LF]	☐ Sentry alerts configured [LF]	☐ PostHog dashboards built [PM]
☐ Customer support inbox ready (Zendesk or Plain) [PM]	☐ Refund policy documented [PM]	☐ Terms of Service live [PM]
☐ Privacy Policy live [PM]	☐ Acceptable Use Policy live [PM]	☐ DPA template ready [SE]
☐ Cookie consent banner implemented [LF]	☐ Sub-processor list published [SE]	☐ Security.txt published [SE]

Part X — Appendices

[FILE INVENTORY](#) · [SCHEMA DIFF](#) · [DEPENDENCY MATRIX](#) · [GLOSSARY](#)

Appendix A — Existing Source File Inventory

Complete inventory of the existing codebase, with status. Files marked **BROKEN** have bugs that prevent correct operation. Files marked **MISSING** are referenced by code but do not exist. Files marked **STUB** have placeholder implementations.

File	Status	Notes
app/layout.tsx	OK	Root layout, fonts, metadata
app/globals.css	OK	Tailwind directives, base styles
app/(marketing)/page.tsx	OK	Landing page — strongest part of codebase
app/(marketing)/login/page.tsx	BROKEN	Generic form, no brand context
app/(marketing)/solutions/[slug]/page.tsx	OK	Industry vertical pages
app/(marketing)/privacy/page.tsx	OK	Static privacy policy
app/(marketing)/terms/page.tsx	OK	Static terms of service
app/(marketing)/layout.tsx	MISSING	Every marketing page re-implements nav/footer
app/(app)/layout.tsx	OK	App shell with AppNav
app/(app)/dashboard/page.tsx	OK	Basic dashboard, no charts
app/(app)/reviews/page.tsx	OK	Reviews list with reply UI
app/(app)/requests/page.tsx	BROKEN	Duplicate , missing features
app/(app)/billing/page.tsx	BROKEN	Enterprise plan unreachable (P0 FB-02)
app/(app)/settings/page.tsx	OK	Basic settings form
app/(app)/org-dashboard/page.tsx	BROKEN	Raw table, no charts, no real data
app/onboarding/page.tsx	BROKEN	Connect Facebook 404s (P0 FB-01)
app/vs/birdeye/page.tsx	OK	Comparison page, no nav/footer
app/vs/podium/page.tsx	OK	Comparison page, no nav/footer
app/not-found.tsx	MISSING	Default Next.js 404

File	Status	Notes
app/error.tsx	MISSING	Default Next.js error
app/loading.tsx	MISSING	Default Next.js loading
app/r/[token]/page.tsx	MISSING	Click tracking route — 404 for every SMS/email recipient
app/api/auth/route.ts	BROKEN	Returns 501 in real mode
app/api/auth/active-business/route.ts	OK	Returns active business ID
app/api/businesses/route.ts	OK	List/create businesses
app/api/businesses/[id]/route.ts	OK	Get/update/delete business
app/api/businesses/[id]/connect/google/route.ts	BROKEN	Mock data leak in production path
app/api/businesses/[id]/connect/facebook/route.ts	MISSING	Called by OnboardingForm, 404s
app/api/businesses/[id]/review-requests/route.ts	OK	List/create review requests
app/api/businesses/[id]/review-requests/quick-add/route.ts	OK	Quick add single customer
app/api/businesses/[id]/review-requests/parse-paste/route.ts	OK	Parse pasted customer list
app/api/businesses/[id]/review-requests/upload/route.ts	OK	CSV upload
app/api/businesses/[id]/review-requests/send-batch/route.ts	OK	Batch send, no rate limiting
app/api/reviews/[id]/draft/route.ts	OK	Generate AI draft
app/api/reviews/[id]/reply/route.ts	OK	Post reply to source
app/api/billing/checkout/route.ts	BROKEN	Rejects enterprise plan (P0 FB-02)
app/api/billing/manage/route.ts	BROKEN	Uses businesses[0] not active business (P1 FB-03)
app/api/inngest/route.ts	BROKEN	No signature verification (P0 BB-03)
app/api/dev/trigger/route.ts	BROKEN	Unauthenticated (P0 BB-01)
app/api/webhooks/stripe/route.ts	MISSING	Referenced but never created (P0 BB-02)
app/api/webhooks/twilio/route.ts	MISSING	Inbound SMS + opt-out not handled
app/api/webhooks/resend/route.ts	MISSING	Bounce/complaint not handled
components/AppNav.tsx	OK	App sidebar nav
components/BillingManager.tsx	BROKEN	Calls enterprise plan, backend rejects
components/OnboardingForm.tsx	BROKEN	Facebook connect 404s

File	Status	Notes
components/RequestsManager.tsx	OK	Review request management UI
components/SettingsForm.tsx	OK	Business settings form
components/LiveDemo.tsx	OK	Interactive landing demo
components/ReviewCard.tsx	OK	Single review card
components/BeforeAfter.tsx	OK	Comparison slider
components/TestimonialCarousel.tsx	OK	Dot-nav carousel
components/PricingToggle.tsx	OK	Monthly/annual toggle (violates Prompt Pack)
components/StatCounter.tsx	BROKEN	Shows "Placeholder stat" text
components/ScrollReveal.tsx	OK	IntersectionObserver fade-in
components/LogoCarousel.tsx	BROKEN	Uses 8 emoji as fake logos
lib/auth.ts	OK	Auth helpers
lib/db.ts	OK	Store factory (local vs supabase)
lib/types.ts	OK	Shared TypeScript types
lib/ai/draft-reply.ts	BROKEN	AI_MODEL is non-existent model name (P1 BB-09)
lib/inngest/client.ts	OK	Inngest client
lib/inngest/index.ts	OK	Function registration
lib/inngest/generate-reply-draft.ts	OK	Draft generation function
lib/inngest/poll-reviews.ts	OK	Review polling function
lib/inngest/send-review-request.ts	OK	Request send function
lib/integrations/stripe.ts	BROKEN	No webhook, phantom subscriptions (P0 BB-02)
lib/integrations/twilio.ts	OK	SMS send, no inbound
lib/integrations/resend.ts	OK	Email send, no bounce handling
lib/integrations/google-business-profile.ts	BROKEN	postReply throws, accountId empty (P0 BB-04)
lib/integrations/facebook-graph.ts	BROKEN	postReply throws (P0 BB-05)
lib/integrations/mock-data.ts	OK	Mock data (must be guarded)
lib/parsing/paste-list-parser.ts	OK	Paste-list parser

File	Status	Notes
lib/supabase/server.ts	OK	Supabase server client
lib/supabase/local-store.ts	OK	Local file store (dev)
lib/supabase/supabase-store.ts	BROKEN	signIn throws in real mode (P0 BB-06)
supabase/migrations/0001_init.sql	OK	Initial schema
supabase/migrations/0002_add_organizations.sql	BROKEN	RLS disabled on organizations (P1 BB-07)
scripts/seed.ts	OK	Seed script for dev

Appendix B — Schema Diff (Existing → Proposed)

Tables to ADD (do not exist in current schema):

- audit_logs — SOC2 + GDPR compliance
- webhook_events — Idempotency + debugging
- oauth_tokens — Secure token storage
- org_members — Multi-tenant RBAC
- api_keys — Public API access
- sentiment_analyses — AI analytics cache
- competitor_snapshots — Competitor intel
- brand_voice_profiles — AI voice training
- reply_templates — Reusable reply snippets
- usage_events — Metered billing
- feature_flags — Progressive rollout
- review_threads — Customer-level threading
- inbox_views — Saved inbox filters
- review_assignments — Review assignee tracking
- campaigns + campaign_events — Campaign engine
- reply_approvals — Approval workflow
- widgets — Widget configurations
- agency_clients — Agency client relationships
- white_label_configs — White-label theming

Tables to MODIFY:

- reviews: add draft_reply_text, draft_reply_status, replied_at, replied_by, sentiment_score, topics
- review_requests: add campaign_id, variant_id, clicked_at, converted_at

- subscriptions: change subscription_plan enum to include 'enterprise', 'agency', 'custom'
- businesses: add timezone, industry, gb_location_id, white_label_config_id
- organizations: add plan, stripe_customer_id, trial_expires_at, churn_risk_score

Indexes to ADD (7 missing on hot paths):

- reviews(business_id, created_at DESC) — Inbox pagination
- review_requests(business_id, status, created_at DESC) — Request list
- reviews(business_id, source, external_id) — Dedup
- org_members(user_id) — Auth lookup
- audit_logs(actor_id, created_at DESC) — Audit search
- usage_events(business_id, period, event_type) — Metered billing query
- campaign_events(campaign_id, status) — Campaign analytics

Appendix C — Third-Party Dependency Matrix

Dependency	Current	Purpose	Action
Next.js	14.2.15	Framework	Upgrade to 15.x (App Router stable)
React	18.3.1	UI library	Stay on 18 (15 not stable yet)
TypeScript	5.6.x	Type safety	Upgrade to 5.7
Tailwind	3.4.x	Styling	Upgrade to 4.0 on release
Supabase	2.111.x	Database + Auth	Stay current
Inngest	3.x	Background jobs	Stay current
Stripe	16.x	Payments	Stay current
Twilio	5.x	SMS + WhatsApp	Stay current
Resend	4.x	Email	Stay current
Anthropic SDK	0.30.x	AI	Upgrade to latest
Zod	3.x	Validation	Stay on 3 (4 has breaking changes)
lucide-react	NOT INSTALLED	Icons	INSTALL in Phase 1
framer-motion	NOT INSTALLED	Animation	INSTALL in Phase 1
@radix-ui/*	NOT INSTALLED	Accessibility primitives	INSTALL in Phase 1
recharts	NOT INSTALLED	Charts	INSTALL in Phase 1

Dependency	Current	Purpose	Action
cmdk	NOT INSTALLED	Command palette	INSTALL in Phase 1
react-hook-form	NOT INSTALLED	Forms	INSTALL in Phase 1
@tanstack/react-table	NOT INSTALLED	DataTable	INSTALL in Phase 1
posthog-js	NOT INSTALLED	Analytics	INSTALL in Phase 2
@sentry/nextjs	NOT INSTALLED	Error tracking	INSTALL in Phase 0
@upstash/redis	NOT INSTALLED	Rate limiting	INSTALL in Phase 0

Appendix D — Glossary

Term	Expansion	Definition
ARPU	Average Revenue Per User	Monthly revenue divided by paying customers
CHS	Customer Health Score	Composite score: usage (40%) + NPS (20%) + tickets (20%) + sentiment (10%) + billing (10%)
CSAT	Customer Satisfaction Score	1–5 scale per touchpoint
CES	Customer Effort Score	How easy was it to do business with us
DPA	Data Processing Addendum	Legal contract required for GDPR enterprise sales
DSAR	Data Subject Access Request	GDPR right: user requests all data about them
GBP	Google Business Profile	Formerly Google My Business
Inngest	Background job framework	Serverless durable functions
NPS	Net Promoter Score	-100 to +100, likelihood to recommend
PITR	Point-In-Time Recovery	Database backup to any second within retention window
RACI	Responsible, Accountable, Consulted, Informed	Project responsibility framework
RLS	Row-Level Security	Postgres feature for per-row access control
RPO	Recovery Point Objective	Max acceptable data loss (time)
RTO	Recovery Time Objective	Max acceptable downtime
SLA	Service Level Agreement	Contractual uptime/response commitment
SOC2	Service Organization Control 2	Audit framework for SaaS security
SSO	Single Sign-On	SAML/OIDC enterprise auth

Term	Expansion	Definition
TCPA	Telephone Consumer Protection Act	US law governing SMS marketing
10DLC	10-Digit Long Code	A2P SMS registration required by US carriers
TAM	Total Addressable Market	Total revenue opportunity if 100% market share

Appendix E — Companion Deliverables

- **ReviewReply_Enterprise_Audit_Roadmap.pdf** — this document (~120 pages)
- **ReviewReply_Bug_Feature_Tracker.xlsx** — sortable, filterable tracker with 80+ items (P0/P1/P2/P3, owner, status, sprint)
- **ReviewReply_Landing_Prototype.html** — premium HTML landing page prototype demonstrating \$50k design feel
- **audit-frontend.md** — full frontend audit report (~12k words)
- **audit-backend.md** — full backend audit report (~14k words)
- **audit-product.md** — full product strategy audit (~56k words)
- **audit-design.md** — full design audit report (~22k words)

End of Document

This concludes the ReviewReply Enterprise Audit, Architecture & Roadmap. The next step is Phase 0 — Stabilization. Approve the budget (\$345k, 24 weeks, 4-person team) and the engineering team can begin Week 1. Companion deliverables (XLSX tracker, HTML landing prototype, audit reports) are available in /home/z/my-project/download/.